



Intelligent Security API (People Counting)

Developer Guide

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Chapter 1 Overview

1.1 Introduction

The people counting function developed via plug-ins provides the statistics of line crossing persons or objects in a specified time period. It integrates the functions of enabling people counting or counting children, search data and report, configuring reverse entering alarm, and so on by Intelligent Security API (ISAPI).

1.2 Update History

The update history shows the summary of changes in ISAPI integration according to different products or projects, and the updated time.

Summary of Changes in Version 2.0_June, 2019

Related Product: DS-2CD6845 Series and DS-2XM6825 Series Network Camera in Software Version 5.5.90

1. Extended people counting capability message ***XML_CountingCap*** (related URL: ***/ISAPI/System/Video/inputs/channels/<ID>/counting/capabilities***):
added 9 nodes, i.e., ***<isSupportRegionsDirection>*** (whether supports setting counting direction), ***<isSupportReverseAlarm>*** (whether supports setting reverse entering alarm), ***<ChildFilter>*** (counting children), ***<MisinfoFilter>*** (false alarm filtering), ***<isSupportParameterExport>*** (whether supports one-touch exporting people counting parameters), ***<detectionMode>*** (detection mode), ***<TrajectoryCountFilter>*** (pattern counting filtering), ***<MultiRegions>*** (number of counting regions), and ***<maintenanceModeEnabled>*** (whether to enable maintenance mode);
added a sub node ***<reportFormat>*** (email report format) to the node ***<EmailReport>***;
deleted three nodes, i.e., ***<isSupportMultiRegions>***, ***<isSupportEnterExitThreshold>***, and ***<isSupportPeopleCountThreshold>***.
2. Extended people counting parameter message ***XML_Counting*** (related URL: ***/ISAPI/System/Video/inputs/channels/<ID>/counting***):
added six nodes, i.e., ***<ChildFilter>*** (counting children), ***<MisinfoFilter>*** (false alarm filtering), ***<detectionMode>*** (detection mode), ***<TrajectoryCountFilter>*** (pattern counting filtering), ***<RegionsDirectionList>*** (parameter list of counting direction), and ***<maintenanceModeEnabled>*** (whether to enable maintenance mode);
added the sub nodes ***<reportFormat>*** (email report format) and ***<child>*** (whether to enable OSD for counting children) to the nodes ***<EmailReport>*** and ***<OverlayConfiguration>***, respectively;
deleted a sub node ***<regionID>*** from the node ***<OverlayConfiguration>***.
3. Extended report capability message of people counting statistics ***XML_CountingSearchCap*** (related URL: ***/ISAPI/System/Video/inputs/channels/<ID>/counting/search/capabilities***):

- added a statistics type "all" (number of entered and exited people) to the node **<statisticsType>**;
- added two nodes **<isSupportchild>** (whether supports counting children only) and **<MinTimeInterval>** (minimum time interval).
- 4. Extended search condition message of people counting report **XML_CountingStatisticsDescription** (related URL: **/ISAPI/System/Video/inputs/channels/<ID>/counting/search**):
 - added a statistics type "all" (number of entered and exited people) to the node **<statisticsType>**;
 - added a node **<child>** (whether supports counting children only);
 - deleted a node **<regionID>**.
- 5. Extended configuration capability message of shielded area **XML_ShieldCap** (related URL: **/ISAPI/Intelligent/channels/<ID>/Shield/capabilities**):
 - added an event type "peopleCounting" to the node **<eventType>**.
- 6. Extended uploaded event message of people counting **XML_EventNotificationAlert_PeopleCountingEventMsg**:
 - added a node **<childCounting>** (counting children).
- 7. Added function of one-touch exporting people counting parameters:
GET /ISAPI/System/Video/inputs/channels/<ID>/counting/parameterExport.
- 8. Added function of **Configure Reverse Entering Alarm**.
- 9. Added function of **Enable Counting Children**.
- 10. Added function of **Enable People Counting**.
- 11. Extended arming schedule message **XML_Schedule** and alarm linkage parameter message **XML_EventTrigger** (related URLs: **/ISAPI/Event/schedules/reverseEntrance/<ID>** and **/ISAPI/Event/triggers/<eventType>-<channelID>**):
 - added an alarm or event type "reverseEntrance" to the node **<eventType>**.
- 12. Added six error codes, refer to **Response Codes of Text Protocol** for details: 0x60000085 (DetectionLineOutOfDetectionRegion), 0x60000086 (DetectionRegionError), 0x60000087 (DetectionRegionOutOfCountingRegion), 0x60000088 (PedalAreaError), 0x60000089 (DetectionAreaABError), and 0x6000008a (ABRegionCannotIntersect).

Summary of Changes in Version 2.0_Mar., 2019

Related Products: DS-9600NI Series, DS-7700NI Series, DS-7600NI Series, and DS-8600NI-K Series Network Video Recorder V4.21.000

Added the function of searching data in multiple video input channels:

Get capability: GET **/ISAPI/System/Video/inputs/channels/counting/search/capabilities**;

Search data: POST **/ISAPI/System/Video/inputs/channels/counting/search**.

Summary of Changes in Version 2.0_Dec., 2018

Fixed the bugs in the document and adjusted the document structure.

Summary of Changes in Version 2.0_Dec., 2017

New document.

Chapter 2 Enable People Counting

During the people counting, the number of entered, exited, or passed people (including adult and child) can be counted.

Before You Start

Make sure you have logged in to device.

Steps

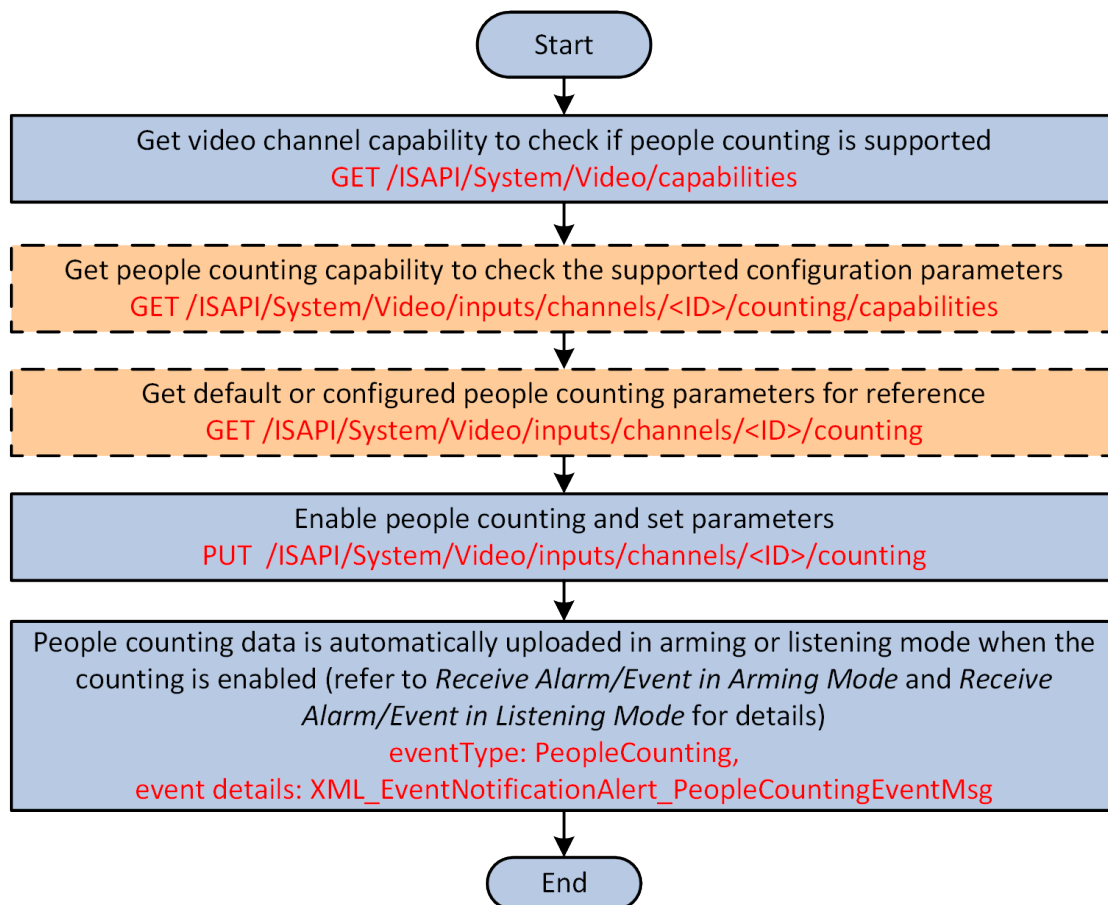


Figure 2-1 Programming Flow of Enabling People Counting

1. Call **/ISAPI/System/Video/capabilities** by GET method to get video channel capability for checking if people counting is supported

The video channel capability is returned in the message **XML_VideoCap**.

If supports, the nodes **<isSupportCounting>** and **<isSupportMultiChannelCounting>** are returned and their values are "true", and then you can perform the following steps; otherwise, it indicates that people counting is not supported, please end this task.

2. **Optional:** Call **/ISAPI/System/Video/inputs/channels/<ID>/counting/capabilities** by GET method to get people counting capability for checking supported configuration parameters.

3. **Optional:** Call ***/ISAPI/System/Video/inputs/channels/<ID>/counting*** by GET method to get default or configured people counting parameters for reference.
4. Call ***/ISAPI/System/Video/inputs/channels/<ID>/counting*** by PUT method and set **<enabled>** of **<Counting>** in message ***XML_Counting*** to "true" for enabling people counting.
5. People counting data is automatically uploaded by the message ***XML_EventNotificationAlert_PeopleCountingEventMsg*** in arming mode (refer to ***Receive Alarm/Event in Arming Mode*** for details) or listening mode (refer to ***Receive Alarm/Event in Listening Mode*** for details) when the counting is enabled.
6. **Optional:** Refer to ***Search for Report or Data*** for searching people counting data and getting report.

Chapter 3 Enable Counting Children

During people counting, the number of entered or exited children can be counted according to the height feature which differs from that of adult for more accurate statistics. The children counting results can also be shown on the people counting report.

Before You Start

Make sure you have logged in to device.

Steps

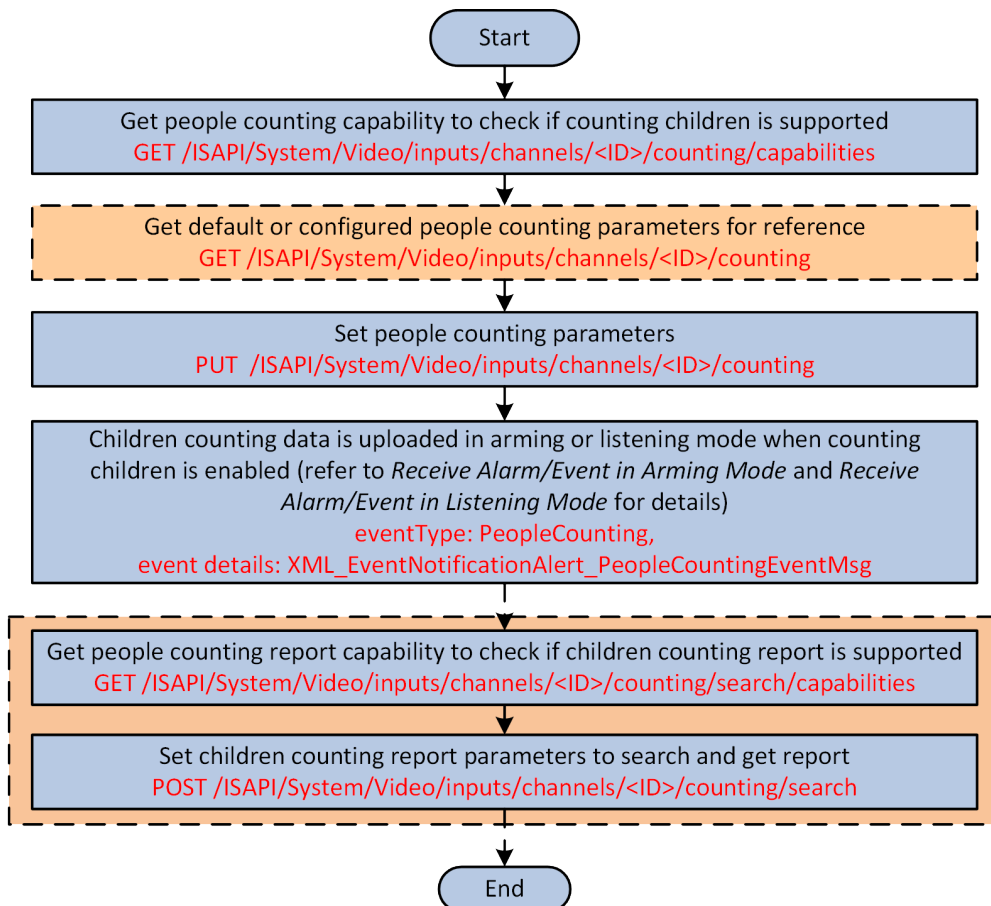


Figure 3-1 Programming Flow of Enabling Counting Children

1. Call `/ISAPI/System/Video/inputs/channels/<ID>/counting/capabilities` by GET method to get people counting capability for checking if counting children is supported.
The people counting capability is returned in the message `XML_CountingCap`.
If supports, the node `<ChildFilter>` is returned, and then you can perform the following steps; otherwise, it indicates that counting children is not supported, please end this task.
2. **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/counting` by GET method to get default or configured people (including children) counting parameters for reference.

3. Call ***/ISAPI/System/Video/inputs/channels/<ID>/counting*** by PUT method and set **<enabled>** of **<ChildFilter>** in message ***XML_Counting*** to "true" for enabling counting children.
4. Children counting data is automatically uploaded by the node **<childCounting>** in the message ***XML_EventNotificationAlert_PeopleCountingEventMsg*** in arming mode (refer to ***Receive Alarm/Event in Arming Mode*** for details) or listening mode (refer to ***Receive Alarm/Event in Listening Mode*** for details) when counting children is enabled.
5. **Optional:** Search and get children counting report.
 - 1) Call ***/ISAPI/System/Video/inputs/channels/<ID>/counting/search/capabilities*** by GET method to get people counting report capability for checking if children counting report is supported.

The people counting report capability is returned in the message ***XML_CountingSearchCap*** . If supports, the node **<isSupportChild>** is returned and its value is "true", and then you can perform the following step; otherwise, it indicates that children counting report is not supported, please end this task.
 - 2) Call ***/ISAPI/System/Video/inputs/channels/<ID>/counting/search*** by POST method and set the search condition to search and get children counting report.

Chapter 4 Search for Report or Data

For devices with local storage, the people counting data will be stored locally after enabling people counting function. You can set condition to search for the stored data and get the different types of reports according to the configured condition.

Table 4-1 Search for People Counting Report or Data

Function	Request URI
Get capability of searching people counting data in a channel	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/counting/search/capabilities</i>
Search people counting data in a channel and get reports	POST <i>/ISAPI/System/Video/inputs/channels/<ID>/counting/search</i>
Get capability of searching people counting data in multiple channels	GET <i>/ISAPI/System/Video/inputs/channels/counting/search/capabilities</i>
Search people counting data in multiple channels and get reports	POST <i>/ISAPI/System/Video/inputs/channels/counting/search</i>

Chapter 5 Alarm or Event Receiving

When the alarm is triggered or the event occurred, if you have configured alarm/event uploading parameters, you can receive and process the alarm/event information in the third-party platform or system. Two modes are available for receiving alarms, including arming mode and listening mode.

Arming Mode

When the alarm is triggered or event occurred, the third-party platform or system can send the request URL to the device for getting the alarm/event stream, and then the device uploads the response message with alarm/event information.

Listening Mode

When alarm is triggered or event occurred, the device uploads the alarm information automatically, and then the third-party platform or system can receives the alarm/event by configuring listening port of HTTP host server.



Note

Currently, for traffic camera or capture camera, receiving alarm or event in arming mode is not supported.

5.1 Configure Reverse Entering Alarm

When a detection target enters the rule region in opposite detection direction, the reverse entering alarm is triggered. You can also set alarm linkage action and arming schedule to receive the alarm information from platform or system and control period that is able to trigger alarm as needed.

Before You Start

Make sure you have logged in to device.

Steps

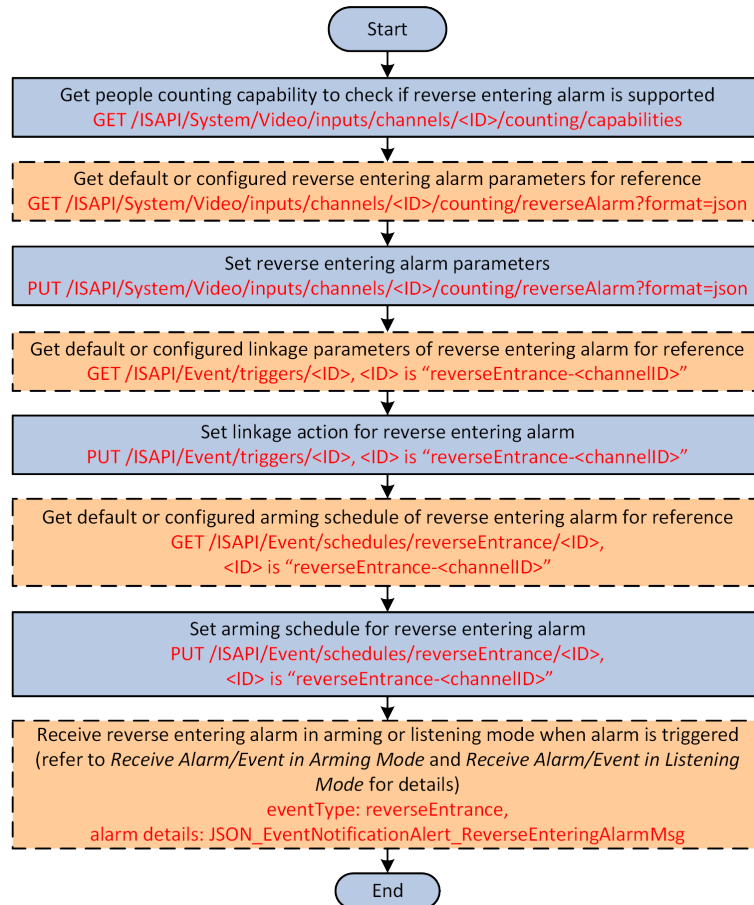


Figure 5-1 Programming Flow of Configuring Reverse Entering Alarm

1. Call **/ISAPI/System/Video/inputs/channels/<ID>/counting/capabilities** by GET method to get people counting capability for checking if reverse entering alarm is supported.
The people counting capability is returned in the message ***XML_CountingCap***.
If supports, the node **<isSupportReverseAlarm>** is returned and its value is "true", and then you can perform the following steps; otherwise, it indicates that reverse entering alarm is not supported, please end this task.
2. **Optional:** Call **/ISAPI/System/Video/inputs/channels/<ID>/counting/reverseAlarm?format=json** by GET method to get default or configured reverse entering alarm parameters for reference.
3. Call **/ISAPI/System/Video/inputs/channels/<ID>/counting/reverseAlarm?format=json** by PUT method to set parameters for reverse entering alarm.
4. **Optional:** Call **/ISAPI/Event/triggers/<eventTypes>-<channelID>** by GET method and set the <ID> in the URL to "reverseEntrance-<channelID>" to get default or configured linkage parameters of reverse entering alarm for reference.
The default linkage action is "center" (alarm receiving center).

5. Call `/ISAPI/Event/triggers/<eventType>-<channelID>` by PUT method and set the <ID> in the URL to "reverseEntrance-<channelID>" to set linkage action for reverse entering alarm.



Note

To receive alarm in alarm receiving center, the linkage action must be set to "center".

6. **Optional:** Call `/ISAPI/Event/schedules/reverseEntrance/<ID>` by GET method and set the <ID> in the URL to "reverseEntrance-<channelID>" to get default or configured arming schedule of reverse entering alarm for reference.
The default arming schedule is "all-day".
7. Call `/ISAPI/Event/schedules/reverseEntrance/<ID>` by PUT method and set the <ID> in the URL to "reverseEntrance-<channelID>" to set arming schedule for reverse entering alarm.
8. **Optional:** Receive reverse entering alarm in arming mode (refer to **Receive Alarm/Event in Arming Mode** for details) or listening mode (refer to **Receive Alarm/Event in Listening Mode** for details).

The reverse entering alarm information is returned in the message

JSON_EventNotificationAlert_ReverseEnteringAlarmMsg .

5.2 Receive Alarm/Event in Arming Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, you can send request message to device for getting the alarm/event stream, and then the device uploads the corresponding response message, which contains alarm/event information.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

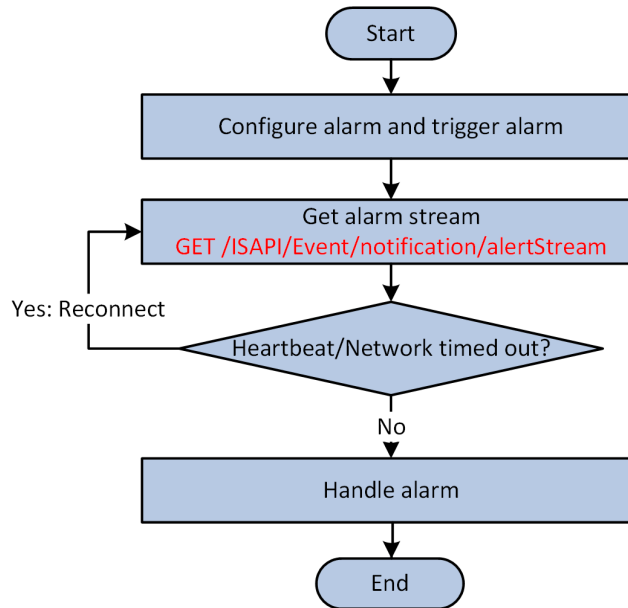


Figure 5-2 API Calling Flow of Receiving Alarm/Event in Arming Mode

1. Call `/ISAPI/Event/notification/alertStream` by GET to get the alarm/event stream.
2. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
3. Receive and process the alarm/event information.

Example

Sample Code of Receiving Alarm/Event in Arming Mode (without Binary Picture Data)

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: data_gateway_ip
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth",
realm="IP Camera (C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",
```

```
realm="IP Camera(C2183)",  
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",  
uri="/ISAPI/Event/notification/alertStream",  
cnonce="3d183a245b8729121ae4ca3d41b90f18",  
nc=00000001,  
qop="auth",  
response="f2e0728991bb031f83df557a8f185178"  
Host: 10.6.165.192
```

```
HTTP/1.1 200 OK  
MIME-Version: 1.0  
Connection: close  
Content-Type: multipart/mixed; boundary=<frontier>
```

```
--<frontier>  
Content-Type: application/xml; charset="UTF-8"  
Content-Length: text_length
```

```
<EventNotificationAlert/>  
--<frontier>
```

5.3 Receive Alarm/Event in Listening Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, the device uploads the alarm/event information automatically, you can receive the alarm/event by configuring the listening port of HTTP host server.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

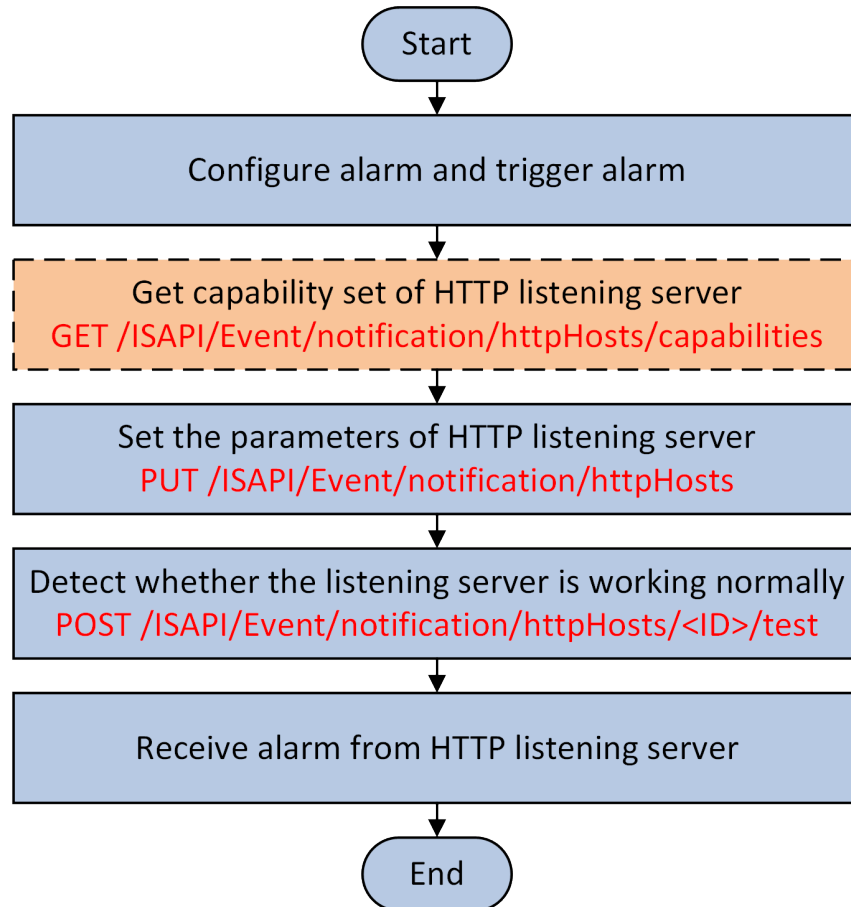


Figure 5-3 API Calling Flow of Receiving Alarm/Event in Listening Mode

1. **Optional:** Call `/ISAPI/Event/notification/httpHosts/capabilities` by GET method to get the capability of HTTP listening server.
2. Call `/ISAPI/Event/notification/httpHosts` by PUT method to set the parameters (including listening address and listening port) of HTTP listening server.

 Note

Before setting the listening server, you'd better call the URI by GET method to get the default or configured parameters for reference.

3. Call `/ISAPI/Event/notification/httpHosts/<ID>/test` by POST method to check if the listening server is working normally.
4. Receive the alarm/event information from the listening server.

 Note

After receiving the alarm/event information uploaded by the device, you need to send a response message with the status "200 OK" to the device as the acknowledgment.

Example

Sample Code of Receiving Alarm/Event in Listening Mode

- with Binary Picture Data

```
//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: body_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml
Content-Length: xml_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[binary picture data]
--<frontier>--

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

- without Binary Picture Data

```
//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: text/xml;
Content-Length: text_length
Connection: keep-alive

<EventNotificationAlert/>

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

Note

- The **Host** is the HTTP server domain name or IP address and port No.
 - Some alarm data is in JSON format, so the **Content-Type** may be "text/xml" or "text/json".
-

Chapter 6 Other Settings

People Counting POS Settings

Function	Request URI
Get the capability of overlaying people counting POS information on the stream of a video input channel	GET <i>/ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay/capabilities</i>
Get parameters for overlaying people counting POS information on the stream of a video input channel	GET <i>/ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay</i>
Set parameters for overlaying people counting POS information on the stream of a video input channel	PUT <i>/ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay</i>

Shielded Area Settings

Function	Request URI
Get configuration capability of shielded area	GET <i>/ISAPI/Intelligent/channels/<ID>/Shield/capabilities</i>
Get or set parameters of a shielded area	GET or PUT <i>/ISAPI/Intelligent/channels/<ID>/Shield/<ID></i>

Advanced Settings

Function	Request URI
Get people counting status	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/counting/status</i>
On-touch export people counting parameters	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/counting/parameterExport</i>
Clear people counting data of local storage	PUT <i>/ISAPI/ContentMgmt/FlashStorage/remove/channels/<ID></i>
Reset people counting statistics	PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/counting/resetCount</i>

Appendix A. Request URIs

A.1 /ISAPI/ContentMgmt/FlashStorage/remove/channels/<ID>

Clear the people counting data in the flash memory.

Request URI Definition

Table A-1 PUT /ISAPI/ContentMgmt/FlashStorage/remove/channels/<ID>

Method	PUT
Description	Clear the people counting data in the flash memory.
Query	None
Request	<i>XML_FlashStorageRemove</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is the video channel ID.

A.2 /ISAPI/Event/schedules/countings

Get or set people counting arming parameters of all channels.

Request URI Definition

Table A-2 GET /ISAPI/Event/schedules/countings

Method	GET
Description	Get people counting arming parameters of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_CountingScheduleList</i> Failed: <i>XML_ResponseStatus</i>

Table A-3 PUT /ISAPI/Event/schedules/countings

Method	PUT
Description	Set people counting parameters arming parameters of all channels.
Query	None.

Request	<i>XML_CountingScheduleList</i>
Response	<i>XML_ResponseStatus</i>

A.3 /ISAPI/Event/schedules/countings/<ID>

Get or set people counting arming parameters of a specific channel.

Request URI Definition

Table A-4 GET /ISAPI/Event/schedules/countings/<ID>

Method	GET
Description	Get people counting arming parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table A-5 PUT /ISAPI/Event/schedules/countings/<ID>

Method	PUT
Description	Set people counting arming parameters of a specific channel.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4 /ISAPI/Event/schedules/reverseEntrance/<ID>

Get or set arming schedule of reverse entering alarm.

Request URI Definition

Table A-6 GET /ISAPI/Event/schedules/reverseEntrance/<ID>

Method	GET
Description	Get arming schedule of reverse entering alarm.

Query	None
Request	None
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table A-7 PUT /ISAPI/Event/schedules/reverseEntrance/<ID>

Method	PUT
Description	Set arming schedule of reverse entering alarm.
Query	None
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the macro definition of event type and channel ID (reverseEntrance-<channelID>, e.g., reverseEntrance-101).

A.5 /ISAPI/Event/triggers/<eventType>-<channelID>

Get, set, or delete the alarm linkage action by channel.

Request URI Definition**Table A-8 GET /ISAPI/Event/triggers/<eventType>-<channelID>**

Method	GET
Description	Get the alarm linkage action by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table A-9 PUT /ISAPI/Event/triggers/<eventType>-<channelID>

Method	PUT
Description	Set the alarm linkage action by channel.
Query	None
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

Table A-10 DELETE /ISAPI/Event/triggers/<eventType>-<channelID>

Method	DELETE
Description	Delete the alarm linkage action by channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <eventType> in the request URI refers to the predefined event or alarm type name, and the <channelID> is the No. of the event detection channel. For example, if the No. of the face capture channel is 101, the "<eventType>-<channelID>" is "faceSnap-101".

A.6 /ISAPI/Intelligent/channels/<ID>/Shield/<ID>

Get or set a shielded area's parameters of an event type by channel.

Request URI Definition

Table A-11 GET /ISAPI/Intelligent/channels/<ID>/Shield/<ID>

Method	GET
Description	Get a shielded area's parameters of an event type.
Query	None
Request	None
Response	Succeeded: <i>XML_Shield</i> Failed: <i>XML_ResponseStatus</i>

Table A-12 PUT /ISAPI/Intelligent/channels/<ID>/Shield/<ID>

Method	PUT
Description	Set a shielded area's parameters of an event type.
Query	None
Request	<i>XML_Shield</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the intelligent channel ID, and the second <ID> is the predefined event/alarm type name.

A.7 /ISAPI/Intelligent/channels/<ID>/Shield/capabilities

Get configuration capability of shielded area by channel.

Request URI Definition

Table A-13 GET /ISAPI/Intelligent/channels/<ID>/Shield/capabilities

Method	GET
Description	Get configuration capability of shielded area.
Query	None
Request	None
Response	Succeeded: <i>XML_ShieldCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the intelligent channel ID.

A.8 /ISAPI/System/Video/capabilities

Get video channel capability.

Request URI Definition

Table A-14 GET /ISAPI/System/Video/capabilities

Method	GET
Description	Get video channel capability.
Query	None
Request	None
Response	Succeeded: <i>XML_VideoCap</i> Failed: <i>XML_ResponseStatus</i>

A.9 /ISAPI/System/Video/inputs/channels/<channelID>/counting/ posInfoOverlay

Get or set parameters for overlaying people counting POS information on the stream of a video input channel.

Request URI Definition

Table A-15 GET /ISAPI/System/Video/inputs/channels/<ID>/counting/posInfoOverlay

Method	GET
Description	Get parameters for overlaying people counting POS information on the stream of a video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PosInfoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table A-16 PUT /ISAPI/System/Video/inputs/channels/<ID>/counting/posInfoOverlay

Method	PUT
Description	Set parameters for overlaying people counting POS information on the stream of a video input channel.
Query	None.
Request	<i>XML_PosInfoOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI is the video channel ID.

A.10 /ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay/capabilities

Get the capability of overlaying people counting POS information on the stream of a video input channel.

Request URI Definition

Table A-17 GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay/capabilities

Method	GET
Description	Get the capability of overlaying people counting POS information on the stream of a video input channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_Cap_PosInfoOverlay</i> Failed: <i>XML_ResponseStatus</i>

A.11 /ISAPI/System/Video/inputs/channels/counting/search

Search for the people counting statistics in a specified time period.

Request URI Definition

Table A-18 POST /ISAPI/System/Video/inputs/channels/counting/search

Method	POST
Description	Search for the people counting statistics in a specified time period.
Query	None
Request	<i>XML_CountingStatisticsDescription</i>
Response	<i>XML_CountingStatisticsResult</i>

A.12 /ISAPI/System/Video/inputs/channels/counting/search/capabilities

Get people counting configuration capability of video input channels.

Request URI Definition

Table A-19 GET /ISAPI/System/Video/inputs/channels/counting/search/capabilities

Method	GET
Description	Get the people counting configuration capability of video input channels.
Query	None
Request	None
Response	Succeeded: <i>XML_CountingSearchCap</i> Failed: <i>XML_ResponseStatus</i>

A.13 /ISAPI/System/Video/inputs/channels/<ID>/counting

Get or set people counting parameters of one video input channel.

Request URI Definitions

Table A-20 GET /ISAPI/System/Video/inputs/channels/<ID>/counting

Method	GET
Description	Get the people counting parameters of one video input channel.
Query	None
Request	None
Response	Succeeded: <i>XML_Counting</i> Failed: <i>XML_ResponseStatus</i>

Table A-21 PUT /ISAPI/System/Video/inputs/channels/<ID>/counting

Method	PUT
Description	Set the people counting parameters of one video input channel.
Query	None
Request	<i>XML_Counting</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The node <ID> in the URI is the video channel ID.

A.14 /ISAPI/System/Video/inputs/channels/<ID>/counting/capabilities

Get capability of people counting statistics.

Request URI Definition

Table A-22 GET /ISAPI/System/Video/inputs/channels/<ID>/counting/capabilities

Method	GET
Description	Get capability of people counting statistics.
Query	None
Request	None
Response	Succeeded: <i>XML_CountingCap</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.15 /ISAPI/System/Video/inputs/channels/<ID>/counting/parameterExport

Export the people counting parameters.

Request URI Definition

Table A-23 GET /ISAPI/System/Video/inputs/channels/<ID>/counting/parameterExport

Method	GET
Description	Export the people counting parameters.
Query	None
Request	None
Response	Succeeded: binary data Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.16 /ISAPI/System/Video/inputs/channels/<ID>/counting/resetCount

Reset the people counting statistics.

Request URI Definition

Table A-24 PUT /ISAPI/System/Video/inputs/channels/<ID>/counting/resetCount

Method	PUT
Description	Reset the people counting statistics.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.17 /ISAPI/System/Video/inputs/channels/<ID>/counting/reverseAlarm?format=json

Get or set parameters of reverse entering alarm.

Request URI Definition

Table A-25 GET /ISAPI/System/Video/inputs/channels/<ID>/counting/reverseAlarm?format=json

Method	GET
Description	Get parameters of reverse entering alarm.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_reverseAlarm</i> Failed: <i>JSON_ResponseStatus</i>

Table A-26 PUT /ISAPI/System/Video/inputs/channels/<ID>/counting/reverseAlarm?format=json

Method	PUT
Description	Set parameters of reverse entering alarm.
Query	format: determine the format of request or response message.
Request	<i>JSON_reverseAlarm</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.18 /ISAPI/System/Video/inputs/channels/<ID>/counting/search

Search for the people counting statistics in a time period.

Request URI Definition

Table A-27 POST /ISAPI/System/Video/inputs/channels/<ID>/counting/search

Method	POST
Description	Search for the people counting statistics in a time period.
Query	None

Request	<i>XML_CountingStatisticsDescription</i>
Response	<i>XML_CountingStatisticsResult</i>

Remarks

The node <ID> in the URI is the video channel ID.

A.19 /ISAPI/System/Video/inputs/channels/<ID>/counting/search/capabilities

Get the people counting configuration capability of a video input channel.

Request URI Definition

Table A-28 GET /ISAPI/System/Video/inputs/channels/<ID>/counting/search/capabilities

Method	GET
Description	Get the people counting configuration capability of a video input channel.
Query	None
Request	None
Response	Succeeded: <i>XML_CountingSearchCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.20 /ISAPI/System/Video/inputs/channels/<ID>/counting/status

Get people counting status by channel.

Request URI Definition

Table A-29 GET /ISAPI/System/Video/inputs/channels/<ID>/counting/status

Method	GET
Description	Get people counting status by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_CountingStatus</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.21 /ISAPI/Event/notification/alertStream

Get the uploaded heartbeat or alarm/event information.

Request URI Definition

Table A-30 GET /ISAPI/Event/notification/alertStream

Method	GET
Description	Get the heartbeat or uploaded alarm/event information.
Query	None.
Request	None.
Response	Option 1: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>XML_EventNotificationAlert_HeartbeatInfo</i> Option 2: <i>JSON_EventNotificationAlert_Alarm/EventInfo</i>  Note The messages here only show the format of alarm/event information to be uploaded. For details, refer to the corresponding alarm/event configuration chapters.

Remarks

- After calling this URI, a persistent connection is set up between the device and the platform, and the alarm or event information will be uploaded from device continuously once the alarm is triggered or event occurred.
- You can check if the XML response message is the heartbeat information according to the nodes <eventType> and <eventState>. If the values of these two node are "videoloss" and "inactive", respectively, the returned message is the heartbeat information.

A.22 /ISAPI/Event/notification/httpHosts

Get or set parameters of all HTTP listening servers, add a HTTP listening server, and delete all HTTP listening servers.

Request URI Definition

Table A-31 GET /ISAPI/Event/notification/httpHosts

Method	GET
Description	Get parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table A-32 PUT /ISAPI/Event/notification/httpHosts

Method	PUT
Description	Set parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotificationList</i>
Response	<i>XML_ResponseStatus</i>

Table A-33 POST /ISAPI/Event/notification/httpHosts

Method	POST
Description	Add a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table A-34 DELETE /ISAPI/Event/notification/httpHosts

Method	DELETE
Description	Delete all HTTP listening servers.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

A.23 /ISAPI/Event/notification/httpHosts/<ID>

Get or set the parameters of a HTTP listening server, and delete a HTTP listening server.

Request URI Definition

Table A-35 GET /ISAPI/Event/notification/httpHosts/<ID>

Method	GET
Description	Get the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotification</i> Failed: <i>XML_ResponseStatus</i>

Table A-36 PUT /ISAPI/Event/notification/httpHosts/<ID>

Method	PUT
Description	Set the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table A-37 DELETE /ISAPI/Event/notification/httpHosts/<ID>

Method	DELETE
Description	Delete a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server No.

A.24 /ISAPI/Event/notification/httpHosts/<ID>/test

Check whether the HTTP listening server is working normally.

Request URI Definition

Table A-38 POST /ISAPI/Event/notification/httpHosts/<ID>/test

Method	POST
Description	Check whether the HTTP listening server is working normally.
Query	None
Request	<i>XML_HttpHostNotification</i>
Response	Succeeded: <i>XML_HttpHostTestResult</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server ID.

A.25 /ISAPI/Event/notification/httpHosts/capabilities

Get the configuration capabilities of all HTTP listening servers.

Request URI Definition

Table A-39 GET /ISAPI/Event/notification/httpHosts/capabilities

Method	GET
Description	Get the configuration capabilities of all HTTP listening servers.
Query	None
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationCap</i> Failed: <i>XML_ResponseStatus</i>

A.26 http://<ipAddress>:<portNo>/<url>

HTTP listening sever sends alarm information to alarm center.

Request URL Definition

Table A-40 POST http://<ipAddress>:<portNo>/<url>

Method	POST
Description	HTTP listening sever sends alarm information to alarm center.
Query	None
Request	None
Response	Succeeded: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>JSON_EventNotificationAlert_Alarm/EventInfo</i> Failed: <i>XML_ResponseStatus</i>

Remarks

- The <ipAddress> in the request URL refers to the IP address or domain name of HTTP listening server, the <portNo> is the port No. of HTTP listening server, and the <url> represents the streaming URL, which is configured via the HTTP listening server.
- The default port No. is 80, so the request URL without port No. is also valid.

Appendix B. Request and Response Messages

B.1 JSON_EventNotificationAlert_ReverseEnteringAlarmMsg

The reverse entering alarm is uploaded in the JSON format of EventNotificationAlert message.

```
Content-Type: multipart/form-data; boundary=MIME_boundary
--MIME_boundary
Content-Type: application/json
Content-Length:

{
  "ipAddress": "",
  "ipv6Address": "",
  "portNo": ,
  "protocol": "",
  "macAddress": "",
  "channelID": ,
  "dateTime": "",
  "activePostCount",
  "eventType": "reverseEntrance",
  /*required, event/alarm type, here it should be set to "reverseEntrance", the
  maximum length is 128 bytes*/
  "eventState": "",
  "eventDescription": ""
  /*refer to the message JSON_EventNotificationAlert_Alarm/EventInfo for
  description details of the above nodes*/
}

--MIME_boundary
Content-Disposition: form-data; name="F4F665D6A18E41308CE9934DCDDD1111";
filename="reverseEntrance.jpg";
Content-Type: image/jpeg
Content-Length: 5798
Content-ID: reverseEntranceImage
```

See Also

JSON_EventNotificationAlert_Alarm/EventInfo

B.2 JSON_ResponseStatus

JSON message about response status

```
{
  "requestURL": "",
  /*optional, string, request URL*/
  "statusCode": ,
```

```
/*optional, int, status code*/
    "statusString": "",
/*optional, string, status description*/
    "subStatusCode": "",
/*optional, string, sub status code*/
    "errorCode": ,
/*required, int, error code, which corresponds to subStatusCode, this field is
required when statusCode is not 1. The returned value is the transformed
decimal number*/
    "errorMsg": "",
/*required, string, error details, this field is required when statusCode is
not 1*/
    "MErrCode": "0xFFFFFFFF",
/*optional, string, error code categorized by functional modules*/
    "MErrDevSelfEx": "0xFFFFFFFF"
/*optional, string, extension of MErrCode. It is used to define the custom
error code, which is categorized by functional modules*/
}
```

Note

- See ***Response Codes of Text Protocol*** for details about the status codes, sub status codes, error codes, and error descriptions.
 - See ***Error Codes Categorized by Functional Modules*** for details about the error codes, error descriptions, and debugging suggestions.
-

B.3 JSON_reverseAlarm

reverseAlarm message in JSON format

```
{
  "reverseAlarm": [{
    "id": "",
/*required, integer32 type, armed region ID*/
    "enabled": ""
/*required, boolean type, whether to enable reverse entering alarm*/
  }]
}
```

B.4 XML_Cap_PosInfoOverlay

XML message about the capability of overlaying people counting POS information on the stream of a video input channel

```
<?xml version="1.0" encoding="UTF-8"?>
<PosInfoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--optional, object, the capability of overlaying people counting POS
information on the stream of a video input channel-->
```

```
<enabled opt="true,false"><!--required, boolean, whether to enable--></enabled>
  <type opt="inOut,inOutPassIdHeight,height"><!--required, string, type:
"inOut", "inOutPassIdHeight", "height"--></type>
</PosInfoOverlay>
```

B.5 XML_Counting

Counting message in XML format

```
<Counting version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--opt-->
    <normalizedScreenWidth><!--req, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <MountingConfiguration><!--opt, mounting parameters-->
    <viewingAngle><!--req, xs: string, "vertical,tilt"--></viewingAngle>
    <mountHeight><!--opt, xs: integer; mounting height, unit: cm--></mountHeight>
    <horizontalDistance><!--opt, xs: integer; horizontal distance, unit: cm--></horizontalDistance>
    <focalLength><!--opt, xs: integer; unit: mm--></focalLength>
  </MountingConfiguration>
  <OverlayConfiguration><!--opt, OSD parameters-->
    <enabled><!--req, xs: boolean--></enabled>
    <positionX><!--req, xs: interger--></positionX>
    <positionY><!--req, xs: interger--></positionY>
    <OSDType>
      <!--dep, it is valid when enabled is "true", xs: string, "enter,
leave,entreLeave,peoplePassing"-->
    </OSDType>
    <child><!--opt, xs: boolean, whether to enable OSD for children counting--></child>
  </OverlayConfiguration>
  <Demarcation><!--opt-->
    <enabled><!--req, xs: boolean--></enabled>
    <DemarcationRegionList><!--req-->
      <DemarcationRegion><!--opt-->
        <id><!--req, xs:integer--></id>
        <RegionCoordinatesList>
          <RegionCoordinates><!--req-->
            <positionX><!--req, xs:integer;coordinate--></positionX>
            <positionY><!--req, xs:integer;coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </DemarcationRegion>
    </DemarcationRegionList>
    <DemarcationLine>
      <StartPoint><!--req-->
```

```
<positionX><!--req, xs: integer--></positionX>
<positionY><!--req, xs: integer--></positionY>
</StartPoint>
<EndPoint><!--req-->
  <positionX><!--req, xs: integer--></positionX>
  <positionY><!-- req, xs: integer--></positionY>
</EndPoint>
</DemarcationLine>
</Demarcation>
<CountingRegionType><!--read-only, req, xs: string, detection type:
"region,line"--></CountingRegionType>
<CountingRegionList><!--opt, detection region list-->
  <CountingRegion><!--opt, detection region-->
    <id><!--req, xs: integer, detection region ID--></id>
    <Direction><!--req, detection direction-->
      <StartPoint><!--req-->
        <positionX><!--req, xs: integer--></positionX>
        <positionY><!--req, xs: integer--></positionY>
      </StartPoint>
      <EndPoint><!--req-->
        <positionX><!--req, xs: integer--></positionX>
        <positionY><!--req, xs: integer--></positionY>
      </EndPoint>
    </Direction>
    <sensitivityLevel><!--req, xs: integer, sensibility--></sensitivityLevel>
    <spaceGenerationSpeed><!--opt, xs: integer--></spaceGenerationSpeed>
    <timeGenerationSpeed><!--opt, xs: integer--></timeGenerationSpeed>
    <countingSpeed><!--opt, xs: integer--></countingSpeed>
    <detectionType><!--opt, xs: string, "auto,head,shoulder"--></
detectionType>
    <objectSizeCorrection><!--opt, xs: integer--></objectSizeCorrection>
    <RegionCoordinatesList><!--req-->
      <RegionCoordinates><!--req-->
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
        <positionY><!--req, xs: integer; y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <RuleRegionCoordinatesList><!--opt, rule frame-->
      <RegionCoordinates><!--req-->
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
        <positionY><!--req, xs: integer; y-coordinate--></positionY>
      </RegionCoordinates>
    </RuleRegionCoordinatesList>
  </CountingRegion>
</CountingRegionList>
<CountingLineItemList><!--opt, detection line list-->
  <CountingLineItem><!--opt, detection line-->
    <id><!--req, xs: integer, detection line ID--></id>
    <direction><!--req, detection line direction-->
      <StartPoint><!--req-->
        <positionX><!--req, xs: integer--></positionX>
        <positionY><!--req, xs: integer--></positionY>
```

```
</StartPoint>
<EndPoint><!--req-->
  <positionX><!--req, xs: integer--></positionX>
  <positionY><!--req, xs: integer--></positionY>
</EndPoint>
</direction>
<sensitivityLevel><!--req, xs: integer, sensitivity--></sensitivityLevel>
<spaceGenerationSpeed><!--opt, xs: integer--></spaceGenerationSpeed>
<timeGenerationSpeed><!--opt, xs: integer--></timeGenerationSpeed>
<countingSpeed><!--opt, xs: integer--></countingSpeed>
<detectionType><!--opt, xs: string, "auto,head,shoulder"--></
detectionType>
<objectSizeCorrection><!--opt, xs: integer--></objectSizeCorrection>
<LineCoordinatesList>
  <Coordinates><!--req-->
    <positionX><!--req, xs: integer; x-coordinate--></positionX>
    <positionY><!--req, xs: integer; y-coordinate--></positionY>
  </Coordinates>
</LineCoordinatesList>
<polyline>
  <!--opt, xs: boolean, this node should be set to "true" when displaying
polyline; otherwise, this node does not exist-->
</polyline>
</CountingLineItem>
</CountingLineItemList>
<dataUploadCycle>
  <!--opt, xs:integer, time period for uploading people counting data-->
</dataUploadCycle>
<SECUploadEnabled>
  <!--opt, xs:boolean, uploading mechanism for each second-->
</SECUploadEnabled>
<InterferenceSuppression><!--opt-->
  <shadow><!--opt, xs:boolean--></shadow>
  <loitering><!--opt, xs:boolean--></loitering>
  <cart><!--opt, xs:boolean--></cart>
</InterferenceSuppression>
<EmailReport><!--opt-->
  <DayReport><!--opt, xs:boolean--></DayReport>
  <WeekReport><!--opt, xs:boolean--></WeekReport>
  <MonthReport><!--opt, xs:boolean--></MonthReport>
  <YearReport><!--opt, xs:boolean--></YearReport>
  <reportFormat opt="excel, csv, txt, xml"><!--opt, xs: string, email report
format, multiple formats can be selected, the default format is EXCEL--></
reportFormat>
</EmailReport>
<CountingCalibrate><!--req-->
  <calibrateType>
    <!--opt ,xs:string, "automatic, manual, no", calibration types, including
Auto, Manual, and None-->
  </calibrateType>
  <SetupParam><!--req, -->
    <height>
```

```
    <!--opt, xs:float, height, unit: centimeter. The default height is 300,
and it is ranging from 200 to 500-->
  </height>
  <tiltAngle>
    <!--opt, ro,xs:float, angle of pitch, unit: degree. The default value
is 0, and it is ranging from 0 to 180-->
  </tiltAngle>
  <heelAngle>
    <!--opt, ro,xs:float, angle of inclination, unit: degree. The default
value is 0, and it is ranging from -90 to 90 -->
  </heelAngle>
</SetupParam>
<CountingArea><!--req,ro,"Counting Region (in Red Frame)"-->
  <RegionCoordinatesList>
    <RegionCoordinates><!--req-->
      <positionX><!--req, xs:integer;coordinate--></positionX>
      <positionY><!--req, xs:integer;coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</CountingArea>
<AutomaticCalib><!--opt,it is valid when calibrateType is "automaticCalib"--
>
  <CalibRegion><!--opt, Calibration Region (in Green Frame)-->
    <RegionCoordinatesList>
      <RegionCoordinates><!--req-->
        <positionX><!-- req, xs:integer;coordinate --></positionX>
        <positionY><!--req, xs:integer;coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </CalibRegion>
</AutomaticCalib>
</CountingCalibrate>
<HeightFilterOverlay><!--opt-->
  <enabled>
    <!--opt, xs:boolean, whether to enable height filter. By default, it is
disabled-->
  </enabled>
  <heightFilter>
    <!--opt,xs:integer, filter height, unit: centimeter. The default value is
120, and it is ranging from 0 to 250-->
  </heightFilter>
</HeightFilterOverlay>
<countingType><!--opt, xs: string, method to trigger counting:
"none,alarmInputTrigger,videoTrigger"--></countingType>
  <signalType>
    <!--dep, xs: string, signal type: "level,pulses", this node is valid when
<b>countingType</b> is set to "videoTrigger" or "alarmInputTrigger"-->
  </signalType>
  <RS485TransmissionEnabled><!--dep, xs: boolean, "false,true"--></
RS485TransmissionEnabled>
  <dailyResetTime>
    <!--opt; xs: ios_8601_time, integer hour, e.g., "22:00:00+08:00"-->
```

```
</dailyResetTime>
<streamOverlayRuleInfos><!--req, xs: boolean, "true,false", whether to enable
overlying rule information in stream--></streamOverlayRuleInfos>
<detectThreshold><!--opt, xs: integer, global detection threshold, value
range: [0,100], default: 90--></detectThreshold>
<aidedTrackEnabled><!--opt, xs: boolean, "true,false", whether to enable
auxiliary tracking, by default, it is enabled--></aidedTrackEnabled>
<TargetRegionCoordinatesList><!--opt, target frame, only supports square-->
  <RegionCoordinates><!--req-->
    <positionX><!--req, xs: integer; x-coordinate--></positionX>
    <positionY><!--req, xs: integer; y-coordinate--></positionY>
  </RegionCoordinates>
</TargetRegionCoordinatesList>
<ChildFilter><!--opt, counting children-->
  <enabled><!--req, xs: boolean, whether to enable counting children: true,
false--></enabled>
  <heightThreshold>
    <!--dep, xs: integer, height threshold of children, the counting will
start when the height of children is higher than configured height threshold,
unit: cm, value range: [100,200]-->
  </heightThreshold>
</ChildFilter>
<MisinfoFilter><!--opt, false alarm filtering-->
  <enabled><!--req, xs: boolean, whether to enable false alarm filtering:
true, false--></enabled>
  <threshold><!--opt, xs: integer, threshold of false alarm filtering, value
range: [0,100], default value: 15, larger threshold value corresponds to higher
precision--></threshold>
  <timesLimit><!--opt,xs:integer, judge times of false alarm, value range:
[1,5], default value: 3, more attempts corresponds to higher precision--></
timesLimit>
</MisinfoFilter>
<detectionMode>
  <!--opt, xs: string, detection mode: single-detect based on tracking
algorithm, double-detect based on depth map, double_single-detect based on
depth map mainly and tracking algorithm secondarily, single_double-detect based
on tracking algorithm mainly and depth map secondarily-->
</detectionMode>
<TrajectoryCountFilter><!--opt, pattern counting filtering-->
  <enabled><!--req, xs: boolean,whether to enable, by default, it is enabled--
--></enabled>
  <movementDisplacement><!--opt, xs: integer, displacement range: [0,200],
default value: 40, unit: cm--></movementDisplacement>
  <residenceTime><!--opt, xs: float, dwell time duration, range: [0,10.0],
default value: 0.1, unit: second--></residenceTime>
</TrajectoryCountFilter>
<RegionsDirectionList><!--opt, counting direction-->
  <RegionsDirection>
    <id><!--req, xs: integer, detection region ID--></id>
    <StartRegionCoordinatesList><!--req, start region-->
      <RegionCoordinates>
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
```

```
<positionY><!--req, xs: integer; y-coordinate--></positionY>
</RegionCoordinates>
</StartRegionCoordinatesList>
<EndRegionCoordinatesList><!--req, end region-->
  <RegionCoordinates><!--req-->
    <positionX><!--req, xs: integer; x-coordinate--></positionX>
    <positionY><!--req, xs: integer; y-coordinate--></positionY>
  </RegionCoordinates>
</EndRegionCoordinatesList>
</RegionsDirection>
</RegionsDirectionList>
<maintenanceModeEnabled><!--opt, xs: boolean, whether to enable maintenance
mode--></maintenanceModeEnabled>
</Counting>
```

B.6 XML_CountingCap

CountingCap message in XML format

```
<CountingCap version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <CountingRegionType><!--opt, xs: string, "line"--></CountingRegionType>
  <dataUploadCycle opt="1,5,10,15,20,30,60"><!--opt, xs: integer--></
dataUploadCycle>
  <isSupportSECUpload><!--opt, xs: boolean--></isSupportSECUpload>
  <isSupportRecommendValue><!--opt, xs: boolean, "true,false"--></
isSupportRecommendValue>
  <isSupportFlashRemoveCouting><!--opt, xs: boolean, "true"--></
isSupportFlashRemoveCouting>
  <OSDType opt="enter,leave,enterLeave,
none,enterPeoplePassing,pepplePassing"><!--opt, xs: string--></OSDType>
  <InterferenceSuppression><!--opt-->
    <shadow opt="true,false"><!--opt, xs: boolean--></shadow>
    <loitering opt="true,false"><!--opt, xs: boolean--></loitering>
    <cart opt="true,false"><!--opt, xs: boolean--></cart>
  </InterferenceSuppression>
  <EmailReport><!--opt-->
    <DayReport opt="true,false"><!--opt, xs: boolean--></DayReport>
    <WeekReport opt="true,false"><!--opt, xs: boolean--></WeekReport>
    <MonthReport opt="true,false"><!--opt, xs: boolean--></MonthReport>
    <YearReport opt="true,false"><!--opt, xs: boolean--></YearReport>
    <reportFormat opt="excel,csv,txt,xml"><!--opt, xs: string, email report
format, multiple formats can be selected, the default format is EXCEL--></
reportFormat>
  </EmailReport>
  <CountingCalibrate><!--req-->
    <calibrateType opt="automatic>manual,no,smart"><!--opt, xs: string--></
calibrateType>
    <SetupParam><!--req-->
      <height min="" max=""><!--opt, xs: float, unit: cm--></height>
      <tiltAngle min="" max=""><!--opt, read-only, xs: float--></tiltAngle>
```

```
<heelAngle min="" max=""><!--opt, read-only, xs: float--></heelAngle>
</SetupParam>
<CountingArea><!--req, read-only, counting area (red frame)-->
  <RegionCoordinatesList size="">
    <RegionCoordinates><!--req-->
      <positionX><!--req, xs: integer; x-coordinate--></positionX>
      <positionY><!--req, xs: integer; y-coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</CountingArea>
<AutomaticCalib><!--opt, dep, this node is valid when <calibrateType> is
set to "automatic"-->
  <CalibRegion><!--opt, calibration area (green frame)-->
    <RegionCoordinatesList size="">
      <RegionCoordinates><!--req-->
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
        <positionY><!--req, xs: integer; y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </CalibRegion>
</AutomaticCalib>
</CountingCalibrate>
<HeightFilterOverlay><!--opt-->
  <enabled opt="true,false" default=""><!--opt, xs: boolean--></enabled>
  <heightFilter min="" max="" default=""><!--opt, xs: integer--></
heightFilter>
</HeightFilterOverlay>
<isSupportCalibrate opt="true,false"><!--opt, xs: boolean--></
isSupportCalibrate>
<isSupportPosInfoOverlay opt="true,false"><!--opt, xs: boolean--></
isSupportPosInfoOverlay>
<countingType opt="none,alarmInputTrigger,videoTrigger"/><!--opt, xs: string,
method to trigger counting-->
<alarmInputTrigger_signalType opt="level,pulses"/><!--opt, xs: string-->
<videoTrigger_RS485Transmission opt="true,false" /><!--opt,xs:boolean-->
<isSupportCountingStatus><!--opt, xs: boolean--></isSupportCountingStatus>
<isSupportTriggerPeopleCountingDataSearch></
isSupportTriggerPeopleCountingDataSearch>
<isSupportDailyResetTime><!--opt, whether supports resetting and clearing by
schedule--></isSupportDailyResetTime>
<Polyline><!--opt, whether supports polyline-->
  <maxPointNumber><!--req,xs:integer--></maxPointNumber>
</Polyline>
<isSupportBusDetectionArea><!--opt, whether supports vehicle-mounted pedal
detection--></isSupportBusDetectionArea>
<isSupportStreamOverlayRuleInfos><!--opt, whether supports overlaying
intelligent information in stream--></isSupportStreamOverlayRuleInfos>
<isSupportMobileDescribeChange><!--opt, whether supports editing the
displayed parameter name of people counting camera--></
isSupportMobileDescribeChange>
<isSupportAdvanceConfiguration><!--opt, whether supports advanced
configuration--></isSupportAdvanceConfiguration>
```

```
<isSupportSearchDoorStates><!--opt, whether supports detecting door status
triggered by signal of vehicle-mounted device--></isSupportSearchDoorStates>
<isSupportRuleRegion><!--opt, xs: boolean, whether supports setting counting
area--></isSupportRuleRegion>
<notSupportStatisticType opt="enternum,exitnum,peoplePassing">
  <!--dep, xs: string, this node is valid when
<isSupportTriggerPeopleCountingDataSearch> is set to "true"-->
</notSupportStatisticType>
<detectThreshold min="0" max="100" def="90"><!--opt, xs: integer, detection
threshold, range: [0,100] default value: 90--></detectThreshold>
<aidedTrackEnabled opt="true,false" def="true"><!--opt, xs: boolean, whether
to enable auxiliary tracking, by default, it is enabled--></aidedTrackEnabled>
<SceneLimitCapList size="2"><!--opt, capability list of scene limit-->
  <SceneLimitCap><!--opt, capability of a scene limit-->
    <minTargetWidth><!--opt, xs: integer, normalized minimum width of target--
></minTargetWidth>
    <maxTargetWidth><!--opt, xs: integer, normalized maximum width of target--
></maxTargetWidth>
    <maxRuleRegionArea><!--opt, xs: integer, normalized maximum area of rule
frame--></maxRuleRegionArea>
  </SceneLimitCap>
</SceneLimitCapList>
<targetRegionEdgesNum min="4" max="4"><!--opt, capability of frame edge
quantity of targetregion--></targetRegionEdgesNum>
<ruleRegionEdgesNum min="4" max="10">
  <!--dep, capability of edge quantity of rule frame, this node is valid when
<isSupportRuleRegion> is set to true-->
</ruleRegionEdgesNum>
<ChildFilter><!--opt, counting children-->
  <enabled opt="true,false"><!--req, xs: boolean, whether to enable counting
children--></enabled>
  <heightThreshold min="100" max="200" def="150">
    <!--dep, xs: integer, height threshold of children, the counting will
start when the height of children is higher than configured height threshold,
unit: cm, value range: [100,200]-->
  </heightThreshold>
</ChildFilter>
<MisinfoFilter><!--opt, false alarm prevention-->
  <enabled opt="true,false"><!--req, xs: boolean, whether to enable false
alarm filtering--></enabled>
  <threshold min="0" max="100" def="15">
    <!--opt, xs: integer, filtering threshold, value range: [0,100], default
value: 15, larger threshold value corresponds to higher precision-->
  </threshold>
  <timesLimit min="1" max="5" def="3">
    <!--opt, xs: integer, prevention attempts of false alarm, value range:
[1,5], default value: 3, more attempts corresponds to higher precision-->
  </timesLimit>
</MisinfoFilter>
<isSupportReverseAlarm><!--opt, xs: boolean, whether supports setting reverse
counting alarm--></isSupportReverseAlarm>
<detectionMode opt="single,double,double_single,single_double"
```

```
def="double_single">
  <!--opt, xs: string, detection mode: single-detect based on tracking
algorithm, double-detect based on depth map, double_single-detect based on
depth map mainly and tracking algorithm secondarily, single_double-detect based
on tracking algorithm mainly and depth map secondarily-->
  </detectionMode>
  <TrajectoryCountFilter><!--opt, pattern counting filtering-->
    <enabled opt="true,false"><!--req, xs: boolean, whether to enable, by
default, it is enabled--></enabled>
    <movementDisplacement min="0" max="200" def="40"><!--opt, xs: integer,
motion displacement range: [0,200], default value: 40, unit: cm--></
movementDisplacement>
    <residenceTime min="0" max="10.0" def="0.1"><!--opt, xs: float, dwell time
duration, range: [0,10.0], default value: 0.1, unit: second--></residenceTime>
  </TrajectoryCountFilter>
  <isSupportParameterExport><!--opt, xs: boolean, whether supports one-touch
exporting people counting parameters--></isSupportParameterExport>
  <MultiRegions min="" max="">
    <!--opt, xs: integer, number of counting regions, this node will not be
returned if people counting in multiple regions is not supported-->
  </MultiRegions>
  <isSupportRegionsDirection><!--opt, xs: boolean, whether supports setting
counting direction--></isSupportRegionsDirection>
  <maintenanceModeEnabled opt="true,false"><!--opt, xs: boolean, whether to
enable maintenance mode--></maintenanceModeEnabled>
</CountingCap>
```

B.7 XML_CountingScheduleList

Message about people counting arming parameters of all channels in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<CountingScheduleListScheduleList version="2.0" xmlns="http://www.isapi.org/
ver20/XMLSchema">
  <Schedule/><!--optional, see details in XML_Schedule-->
</CountingScheduleListScheduleList>
```

See Also

XML_Schedule

B.8 XML_CountingSearchCap

CountingSearchCap message in XML format

```
<CountingSearchCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <statisticType opt="enternum,exitnum,peoplePassing,all">
    <!--req, xs: string, enternum: number of entered people, exitnum: number of
exited people, peoplePassing: number of passed people, all: number of entered
```

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and exited people-->
</statisticType>
<reportType opt="daily,weekly,monthly,yearly">
  <!--req, xs: string-->
</reportType>
<timeSpanList size="">
  <timeSpan>
    <startTime><!--req, xs: datetime--></startTime>
    <endTime><!--req, xs: datetime--></endTime>
  </timeSpan>
</timeSpanList>
<triggerPeopleCountingData opt="true,false">
  <!--opt, xs: boolean, whether triggers people counting statistics-->
</triggerPeopleCountingData>
<MinTimeInterval opt="quarter, half, hour, day, month, week"><!--opt, xs:
string, minimum time interval--></MinTimeInterval>
<regionsID opt="1,2,3.."><!--opt, xs: integer, region ID, which supports
report format--></regionsID>
<isSupportChild><!--opt,xs: boolean, whether supports counting children only--
></isSupportChild>
</CountingSearchCap>
```

B.9 XML_CountingStatisticsDescription

CountingStatisticsDescription message in XML format

```
<CountingStatisticsDescription version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <statisticType>
    <!--req, xs:string, "enternum,exitnum,peoplePassing,all"-->
  </statisticType>
  <reportType>
    <!--req, xs:string, "daily,weekly,monthly,yearly"-->
  </reportType>
  <timeSpanList>
    <timeSpan>
      <startTime><!--req, xs:datetime--></startTime>
      <endTime><!--req, xs:datetime--></endTime>
    </timeSpan>
  </timeSpanList>
  <triggerPeopleCountingData><!--opt, xs: boolean, whether to enable people
counting: true, false--></triggerPeopleCountingData>
  <PagingSearchCond><!--dep, this node is valid when
<triggerPeopleCountingData> is set to true-->
    <searchID><!--req, xs: string, search ID, which is used to save the search
history for fast search at next time--></searchID>
    <searchResultPosition><!--req, xs: integer--></searchResultPosition>
    <maxResults><!--req, xs: integer --></maxResults>
  </PagingSearchCond>
  <MinTimeInterval><!--opt, xs: string, minimum time interval: quarter, half,
```

```
hour, day, month, week--></MinTimeInterval>
  <child><!--opt, xs: boolean, whether supports counting children only, true:
yes, false: no (count adult and child)--></child>
</CountingStatisticsDescription>
```

B.10 XML_CountingStatisticsResult

CountingStatisticsResult message in XML format

```
<CountingStatisticsResult version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <responseStatus><!--req, xs:string--></responseStatus>
  <responseStatusStrg>
    <!--req, xs:string: "NO MATCHES"-no match, "MORE"-more items to be
searched, "OK"-search completed-->
  </responseStatusStrg>
  <numOfMatches><!--req, xs:integer--></numOfMatches>
  <matchList><!--opt-->
    <matchElement><!--opt-->
      <timeSpan><!--opt-->
        <startTime><!--req, xs:datetime --></startTime>
        <endTime><!--req, xs:datetime --></endTime>
      </timeSpan>
      <enterCount><!--dep, xs:integer--></enterCount>
      <exitCount><!--dep, xs:integer--></exitCount>
      <peoplePassingCount><!--opt, xs:integer--></peoplePassingCount>
    </matchElement>
  </matchList>
</CountingStatisticsResult>
```

B.11 XML_CountingStatus

CountingStatus message in XML format

```
<CountingStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status><!--req, xs: string, "counting,stopped,paused"--></status>
  <time><!--req, operation time (local time + time zone), e.g.,
"22:00:00+08:00"--></time>
  <doorStatus><!--opt, xs: string, door triggering status: "open,close,N_A"--></
doorStatus>
</CountingStatus>
```

B.12 XML_EventNotificationAlert_PeopleCountingEventMsg

XML message about people counting results

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transmission protocol type: "HTTP",
"HTTPS", "EHome"--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <relatedChannelList>
    <!--optional, xs:string, list of alarm related channels, which are of the
same camera with channelID; this parameter is used for live view or playback on
the platform; multiple channel numbers are separated by commas-->
  </relatedChannelList>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601
format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></
activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to
"PeopleCounting"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event):
"active", "inactive"--></eventState>
  <eventDescription><!--required, xs:string, event description--></
eventDescription>
  <channelName><!--optional, xs: string, channel name--></channelName>
  <deviceID><!--optional, xs: string, device ID--></deviceID>
  <peopleCounting>
    <statisticalMethods>
      <!--optional, xs:string, people counting method: "realTime"-real-time
people counting, "timeRange"-scheduled people counting, "signalTrigger"-alarm
input triggered people counting-->
    </statisticalMethods>
    <RealTime>
      <!--dependent, real-time people counting time, this node is required when
statisticalMethods is "realTime"-->
      <time><!--required, xs:string, people counting time--></time>
    </RealTime>
    <TimeRange>
      <!--dependent, time period for real-time people counting, this node is
required when statisticalMethods is "timeRange" or "signalTrigger"-->
      <startTime><!--required, xs:string, start time of people counting--></
startTime>
      <endTime><!--required, xs:string, end time of people counting--></endTime>
    </TimeRange>
    <enter><!--optional, xs:integer, number of entered people--></enter>
    <exit><!--optional, xs:integer, number of exited people--></exit>
    <pass><!--optional, xs:integer, number of passed people--></pass>
    <duplicatePeople><!--optional, xs:integer, number of duplicated people--></
duplicatePeople>
```

```
</peopleCounting>
<childCounting>
  <enter><!--optional, xs: integer, number of entered children--></enter>
  <exit><!--optional, xs: integer, number of exited children--></exit>
</childCounting>
<FaceExpressionList><!--optional, face expression statistics-->
  <FaceExpression><!--list-->
    <value>
      <!--required, xs:string, face expression types: "unknown, poker-faced,
happy, surprised, disgusted, sad, angry, contemptuous, panic, all"-->
    </value>
    <count><!--required, xs:integer, number of people corresponding to the
face expression--></count>
  </FaceExpression>
</FaceExpressionList>
<GenderList><!--optional, gender statistics-->
  <Gender><!--list-->
    <value><!--required, xs:string, gender: "unknown", "female", "male",
"all"--></value>
    <count><!--required, xs:integer, number of people of the corresponding
gender--></count>
  </Gender>
</GenderList>
<MaskList><!--optional, mask statistics-->
  <Mask><!--list-->
    <value><!--required, xs:string, whether the person is wearing a mask:
"unknown, yes, no, all"--></value>
    <count><!--required, xs:integer, number of detected people--></count>
  </Mask>
</MaskList>
<AgeGroupList><!--optional, age group statistics-->
  <AgeGroup><!--list-->
    <value>
      <!--required, xs:string, age group: "unknown, child, young, middle,
old, infant, kid, teenager, prime, middleAged, all"-->
    </value>
    <count><!--required, xs:integer, number of detected people--></count>
  </AgeGroup>
</AgeGroupList>
<GlassList><!--optional, glasses statistics-->
  <Glass><!--list-->
    <value><!--required, xs:string, whether the person is wearing glasses:
"unknown, yes, no, sunglasses, all"--></value>
    <count><!--required, xs:integer, number of detected people--></count>
  </Glass>
</GlassList>
<isDataRetransmission><!--opt, xs:boolean, data retransmission mark--></
isDataRetransmission>
  <targetAttrs><!--optional, transparent transmission field, the following
fields are recommended-->
    <![CDATA[{
/*optional, transparent transmission field, the following fields are
```

```

recommended*/
    "deviceId": "",
/*optional, string, device ID, the maximum size is 64 bytes*/
    "deviceChannel": ,
/*optional, integer32, device channel ID*/
    "deviceName": ""
/*optional, string, device name, the maximum size is 128 bytes*/
  ]]]>
</targetAttrs>
<TaskInfo><!--optional, task information submitted by Data Center products-->
  <taskID><!--required, xs:string, task ID, the maximum size is 64 bytes--></
taskID>
  <algorithmType>
    <!--optional, xs:integer, algorithm type: 1-"hightQualityVehicle" (high-
quality vehicle), 2-"hightPerformanceVehicle" (high-performance vehicle),
3-"personStructModel" (person structural modeling),
4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),
5-"videoFaceDetect" (video face detection), 6-"videoFaceDetectModel" (video
face detection and modeling), 7-"smart" (face detection (output frame without
recognizing properties)+human body detection (output properties and frame)
+behavior detection)-->
  </algorithmType>
  <taskName><!--optional, xs:string, task name--></taskName>
  <algorithmID><!--optional, xs:string, algorithm package ID--></algorithmID>
  <monitorPointID><!--optional, xs:string, camera No.--></monitorPointID>
  <monitorPointName><!--optional, xs:string, camera name--></monitorPointName>
  <streamType><!--optional, xs:string, stream type:
"realtime,historyvideo,localvideo--></streamType>
  <ruleID><!--optional, xs:string, rule ID, which is used to distinguish
different tasks with the same name, and the ruleID in the event uploading
message indicates that the event is related to this task--></ruleID>
  <ruleCustomName><!--required, xs:string, custom event name, which is
included in the message when the alarm is uploaded, and the maximum size is 128
bytes--></ruleCustomName>
  <algorithmInfo><!--optional, algorithm version information, this field is
required for secondary analysis-->
    <algorithmId><!--optional, xs:string, algorithm package --></algorithmId>
    <algoVendorId><!--optional, xs:string, manufacturer ID--></algoVendorId>
    <algoVersionId><!--optional, xs:string, algorithm version ID--></
algoVersionId>
    <algoChipType><!--optional, xs:string, chip type--></algoChipType>
    <algoModelVersion><!--optional, xs:string, model version--></
algoModelVersion>
  </algorithmInfo>
</TaskInfo>
<bkgUrl><!--optional, xs:string, background picture URL--></bkgUrl>
<timeStamp><!--optional, xs:string, time of the alarm being triggered, ISO
8601 time format--></timeStamp>
<DetectionRegionList><!--optional, detected area-->
  <DetectionRegionEntry><!--list-->
    <regionID><!--required, xs:string --></regionID>
    <sensitivityLevel><!--optional, xs:integer, the value ranges from 0

```

```
to 100 --></sensitivityLevel>
    <RegionCoordinatesList><!--optional, target area-->
        <RegionCoordinates>
            <!--optional-->
                <positionX><!--required, xs:integer; coordinate, [0,1000]--></
positionX>
                <positionY><!--required, xs:integer; coordinate, [0,1000]--></
positionY>
            </RegionCoordinates>
        </RegionCoordinatesList>
    </DetectionRegionEntry>
</DetectionRegionList>
</EventNotificationAlert>
```

Example

Sample Message of People Counting Results in XML Format

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
    <ipAddress>10.10.34.138</ipAddress>
    <protocolType>HTTP</protocolType>
    <macAddress>00:40:30:11:11:12</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-08-25T11:34:59+08:00</dateTime>
    <activePostCount>3</activePostCount>
    <eventType>PeopleCounting</eventType>
    <eventState>active</eventState>
    <eventDescription>peopleCounting alarm</eventDescription>
    <channelName>Camera 01</channelName>
    <peopleCounting>
        <statisticalMethods>realTime</statisticalMethods>
        <RealTime>
            <time>2017-08-25T11:34:59+08:00</time>
        </RealTime>
        <enter>34</enter>
        <exit>39</exit>
        <pass>0</pass>
    </peopleCounting>
</EventNotificationAlert>
```

B.13 XML_EventTrigger

Linkage parameter message in XML format

```
<EventTrigger version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <id><!--required, xs:string, ID--></id>
    <eventType>
        <!--required, xs:string, see details in the "Remarks" below-->
    </eventType>
    <eventDescription><!--optional, xs:string--></eventDescription>
```

```
<inputIOPortID><!--dependent, xs:string, alarm input ID--></inputIOPortID>
<dynInputIOPortID><!--dependent, xs:string, dynamic alarm input ID--></
dynInputPortID>
<videoInputChannelID>
  <!--dependent, xs:string, video input channel ID, it is valid when
<eventType> is "VMD, videoloss, tamperdetection, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage"-->
</videoInputChannelID>
<dynVideoInputChannelID><!--dependent, xs:string, dynamic video input channel
ID--></dynVideoInputChannelID>
<intervalBetweenEvents><!--optional, xs:integer, event time interval, unit:
second--></intervalBetweenEvents>
<WLSensorID><!--dependent, xs:string, ID--></WLSensorID>
<EventTriggerNotificationList/><!--optional, alarm/event linkage actions, see
details in the message of XML_EventTriggerNotificationList-->
</EventTrigger>
```

Remarks

The node **<eventType>** can be the following values: IO, VMD, videoloss, raidfailure, recordingfailure, badvideo, POS, analytics, fanfailure, overheat, tamperdetection, diskfull, diskerror, nicbroken, ipconflict, illaccess, videomismatch, resolutionmismatch, radifailure, PIR, WLSensor, spareException, poePowerException, heatmap, counting, linedetection, fielddetection, regionEntrance, regionExiting, loitering, group,rapidMove, parking, unattendedBaggage, attendedBaggage, HUMANATTRIBUTE, blacklist, whitelist, peopleDetection, allVehicleList, otherVehicleList, vehicledetection, storageDetection, shipsDetection, humanAttribute, faceContrast, blacklistFaceContrast, whitelistFaceContrast, faceSnap, faceLib, personDensityDetection, personQueueDetecton, mixedTargetDetection, HVTVehicleDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, thermometry, shipsFlowDetection, dredgerDetection, reverseEntrance, luma, highHDTemperature, lowHDTemperature, hdImpact, hdBadBlock, SevereHDFailure, safetyHelmetDetection, vibrationDetection, HBDLib,TMPA,faceThermometry,noMaskDetection

See Also

XML_EventTriggerNotificationList

B.14 XML_EventTriggerNotification

Event linkage notification message in XML format

```
<EventTriggerNotification><!--opt-->
  <id><!--required, xs:string, device ID--></id>
  <notificationMethod>
    <!--required, xs:string, linkage actions,
opt="email, IM, IO, syslog, HTTP, FTP, beep, ptz, record, monitorAlarm, center,
LightAudioAlarm, focus, trace, cloud, SMS, whiteLight, audio, whiteLight, faceContrast, s
iren, output"-->
```

```
</notificationMethod>
<notificationRecurrence>
  <!--optional, xs:string, "beginning,beginningandend,recurring"-->
</notificationRecurrence>
<notificationInterval><!--dependent, xs:integer, unit: millisecond--></
notificationInterval>
  <outputIOPortID><!--dependent, xs:string, video output No., it is required
only when notificationMethod is "IO"--></outputIOPortID>
  <dynOutputIOPortID><!--dependent, xs:string, dynamic video output No., it is
required only when notificationMethod is "IO"--></dynOutputIOPortID>
  <videoInputID><!--dependent, xs:string, video input No., it is required only
when notificationMethod is "record"--></videoInputID>
  <dynVideoInputID><!--dependent, xs:string, dynamic video input No., it is
required only when notificationMethod is "record"--></dynVideoInputID>
  <ptzAction><!--dependent, it is required only when notificationMethod is
"ptz"-->
    <ptzChannelID><!--required, xs:string, PTZ channel ID--></ptzChannelID>
    <actionName><!--required, xs:string, PTZ control type: "preset", "pattern",
"patrol"--></actionName>
    <actionNum><!--dependent, xs:integer></actionNum>
  </ptzAction>
  <WhiteLightAction><!--dependent, white light linkage parameters, this node is
valid when notificationMethod is "whiteLight"-->
    <whiteLightDurationTime><!--required, xs:integer, white light flashing
duration, it is between 1 and 60, unit: second--></whiteLightDurationTime>
  </WhiteLightAction>
  <cellphoneNumber><!--dependent, xs:string, min="0" max="11",cellphone number--
></cellphoneNumber-->
</EventTriggerNotification>
```

B.15 XML_EventTriggerNotificationList

EventTriggerNotificationList message in XML format

```
<EventTriggerNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <EventTriggerNotification/><!--opt, see details in the message of
XML_EventTriggerNotification-->
</EventTriggerNotificationList>
```

See Also

XML_EventTriggerNotification

B.16 XML_FlashStorageRemove

FlashStorageRemove message in XML format

```
<FlashStorageRemove version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <coutingRemove>
    <!--opt, xs:boolean-->
  </coutingRemove>
</FlashStorageRemove>
```

B.17 XML_PosInfoOverlay

XML message about parameters for overlaying people counting POS information on the stream of a video input channel

```
<PosInfoOverlay version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <!--optional, object, parameters for overlaying people counting POS
information on the stream of a video input channel-->
  <enabled><!--optional, xs:boolean, whether to enable--></enabled>
  <type><!--required, xs:string, type: "inOut" (entered and exited
information), "inOutPassIdHeight" (entered, exited, passed, person/object No.,
and height information), "height"--></type>
</PosInfoOverlay>
```

B.18 XML_ResponseStatus

XML message about response status

```
<?xml version="1.0" encoding="utf-8"?>
<ResponseStatus version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <requestURL>
    <!--required, read-only, xs:string, request URL-->
  </requestURL>
  <statusCode>
    <!--required, read-only, xs:integer, status code: 0,1-OK, 2-Device Busy, 3-
Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content,
7-Reboot Required, 9-Additional Error-->
  </statusCode>
  <statusString>
    <!--required, read-only, xs:string, status description: OK, Device Busy,
Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot, Additional Error-->
  </statusString>
  <subStatusCode>
    <!--required, read-only, xs:string, describe the error reason in detail-->
  </subStatusCode>
  <MErrCode>
    <!--optional, xs:string, error code categorized by functional modules,
e.g., 0x12345678-->
  </MErrCode>
  <MErrDevSelfEx>
```

```
<!--optional, xs:string, extension field of MErrCode. It is used to define
the custom error code, which is categorized by functional modules-->
</MErrDevSelfEx>
</ResponseStatus>
```

Note

- See **Response Codes of Text Protocol** for details about sub status codes and corresponding error codes.
 - See **Error Codes Categorized by Functional Modules** for details about the error codes, error descriptions, and debugging suggestions.
-

B.19 XML_Schedule

Schedule message in XML format

```
<Schedule version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--optional, xs:string, alarm/event types, see details in the "Remarks"
below-->
  </eventType>
  <inputIOPortID><!--read-only, dependent, xs:string, alarm input No.--></
inputIOPortID>
  <outputIOPortID><!--read-only, dependent, xs:string, alarm output No.--></
inputIOPortID>
  <videoInputChannelID><!--read-only, dependent, xs:string, video input channel
ID--></videoInputChannelID>
  <TimeBlockList size="8"><!--required-->
    <TimeBlock><!--list-->
      <dayOfWeek>
        <!--optional, xs:integer, day of the week based on ISO8601,
"1"=Monday, ...-->
      </dayOfWeek>
      <TimeRange><!--required-->
        <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
        <endTime><!--required, xs:time, ISO8601 time--></endTime>
      </TimeRange>
      <CustomExtension>
        <vehicleDetectSceneID>
          <!--required, xs:interger-->
        </vehicleDetectSceneID>
      </CustomExtension>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList><!--optional-->
    <TimeBlock><!--list-->
      <TimeRange><!--required-->
        <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
        <endTime><!--required, xs:time, ISO8601 time--></endTime>
```

```
</TimeRange>
</TimeBlock>
</HolidayBlockList>
</Schedule>
```

Remarks

The node **<eventType>** can be set to the following values: IO, VMD, videoloss, PIR, linedetection, fielddetection, audioexception, facedetection, regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, storageDetection, shipsDetection, HUMANATTRIBUTE, humanAttribute, faceContrast, faceSnap, faceLib, whiteListFaceContrast, personDensityDetection, personQueueDetection, mixedTargetDetection, fireDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, reverseEntrance, vehicledetect, safetyHelmetDetection, vibrationDetection, TMPA, faceThermometry, HBDLib.

B.20 XML_Shield

Message about shielded area parameters in XML format

```
<Shield version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:string, whether enables the shield function:
"true,false"--></enabled>
  <RegionList>
    <Region>
      <id><!--req, xs:string, area ID--></id>
      <enabled><!--req, xs:string, whether enables the shield function of this
area: "true,false"--></enabled>
      <RegionCoordinatesList>
        <RegionCoordinates><!--req-->
          <positionX><!--req, xs:integer, X-coordinate--></positionX>
          <positionY><!--req, xs:integer, Y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

B.21 XML_ShieldCap

Message about configuration capability of shielded area in XML format

```
<ShieldCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--req, read-only-->
    <normalizedScreenWidth><!--req, read-only, xs: integer--></
normalizedScreenWidth>
    <normalizedScreenHeight><!--req, read-only, xs: integer--></
normalizedScreenHeight>
```

```
</normalizedScreenSize>
<enabled opt="true,false"><!--req, xs: string, whether to enable all shielded
areas--></enabled>
<regionNum max="8"><!--req, xs: integer, number of shielded areas--></
regionNum>
<RegionCap><!--req, shielded area capability-->
  <id min="1" max="8"><!--req, xs: string, area ID--></id>
  <enabled opt="true,false"><!--req, xs: string, whether to enable the
specified shield area--></enabled>
  <RegionCoordinatesList size=""><!--req, shielded area coordinates-->
    <RegionCoordinates><!--req, list-->
      <positionX><!--req, xs: integer, x-coordinate--></positionX>
      <positionY><!--req, xs: integer, y-coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</RegionCap>
<eventType opt="faceCapture,mixedTargetDetection,peopleCounting"><!--opt, xs:
string, event type--></eventType>
</ShieldCap>
```

B.22 XML_VideoCap

VideoCap message in XML format

```
<VideoCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoInputPortNums><!--opt, xs: integer--><videoInputPortNums>
  <videoOutputPortNums><!--opt, xs: integer--><videoOutputPortNums>
  <isSupportHeatmap><!--opt, xs: boolean, whether to support heat map function--
></isSupportHeatmap>
  <isSupportCounting><!--opt, xs: boolean--></isSupportCounting>
  <countingType><!--dep, xs: string, "human,object"--></countingType>
  <isSupportPreviewSwitch><!--opt, xs: boolean, whether to support live view
switch--></isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--opt, xs: boolean, whether supports getting recording status-->
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--opt, xs: boolean, whether supports priavte mask function-->
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--opt, xs: boolean, whether supports auto-switch during the live view
of dual lens camera-->
  </isSupportBinocularPreviewSwitch>
  <isSupportCalibCheck>
    <!--opt, xs: boolean, whether supports calibration verification-->
  </isSupportCalibCheck>
  <isSupportPIP><!--opt, xs: boolean, "true, false"--></isSupportPIP>
  <channelFlexible opt="name,enable,online,linknum">
    <!--capability of getting channel status by condition-->
  </channelFlexible >
```

```
<isSupportFocusVideoMode>
  <!--opt, xs: boolean, video focus mode for installation and debug-->
</isSupportFocusVideoMode>
<isSupportExternalChannel>
  <!--opt, xs: boolean, whether supports extending analog channel-->
</isSupportExternalChannel>
<isSupportMultiChannelCounting>
  <!--opt, xs: boolean, whether supports people counting of multiple
channels-->
</isSupportMultiChannelCounting>
<isSupportCountingCollection>
  <!--opt, xs:boolean, whether supports people counting data replenishment-->
</isSupportCountingCollection>
<isSupportHeatmapCollection>
  <!--opt, xs:boolean, whether supports heat map data replenishment-->
</isSupportHeatmapCollection>
<OSDLanguage opt="GBK,EUC-KR" def="GBK"/>
<isSupportInitLens><!--req, xs:boolean, whether to support initializing lens--
><isSupportInitLens>
  <isSupportOneFocus><!--req, xs:boolean, whether to support one-touch
focusing--><isSupportOneFoucs>
  <notSupportFocus><!--req, xs:boolean, the focus capability is not supported--
><notSupportFoucs>
  <notSupportIris><!--req, xs:boolean, the iris capability is not supported--
><notSupportIris>
  <isSupportCapturePicOverlays><!--opt, xs:boolean, whether to support text
overlay on the captured picture--></isSupportCapturePicOverlays>
  <isSupportMergePicOverlays><!--opt, xs:boolean, whether to support text
overlay on the composite picture--></isSupportMergePicOverlays>
</VideoCap>
```

B.23 JSON_EventNotificationAlert_Alarm/EventInfo

EventNotificationAlert message with alarm or event information in JSON format.

```
{
  "ipAddress": "",
  /*required, device IPv4 address , string, the maximum size is 32 bytes*/
  "ipv6Address": "",
  /*optional, device IPv6 address, string, the maximum size is 128 bytes*/
  "portNo": ,
  /*optional, device port No., integer32*/
  "protocol": "",
  /*optional, protocol type, "HTTP, HTTPS", string, the maximum size is 32 bytes*/
  "macAddress": "",
  /*optional, MAC address, string, the maximum size is 32 bytes, e.g.,
01:17:24:45:D9:F4*/
  "channelID": "",
  /*optional, device channel No., integer32*/
  "dateTime": "",
```

```
/*optional, string, alarm/event triggered or occurred time based on ISO8601,
the maximum size is 32 bytes, e.g., 2009-11-14T15:27Z*/
    "activePostCount": "",
/*required, alarm/event frequency, integer32*/
    "eventType": "",
/*required, alarm/event type, "captureResult, faceCapture,...", string, the
maximum size is 128 bytes*/
    "eventState": "",
/*required, string, the maximum size is 32 bytes, durative alarm/event status:
"active"-valid, "inactive"-invalid*/
    "eventDescription": "",
/*required, event description, string, the maximum size is 128 bytes*/
    "deviceId": "",
/*string type, device ID*/
    "uuid": "",
/*string type, event UUID, which is used to uniquely identify an event, the
standard UUID format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx*/
    ...
/*optional, for different alarm/event types, the nodes are different, see the
message examples in different applications*/
}
```

B.24 XML_EventNotificationAlert_AlarmEventInfo

EventNotificationAlert message with alarm/event information in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
    <ipAddress><!--dep, xs:string, device IPv4 address--></ipAddress>
    <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>
    <portNo><!--opt, xs:integer, device port number--></portNo>
    <protocol><!--opt, xs:string, protocol type for uploading alarm/event
information, "HTTP,HTTPS"--></protocol>
    <macAddress><!--opt, xs:string, MAC address--></macAddress>
    <channelID><!--dep, xs:string, device channel No., starts from 1--></
channelID>
    <dateTime><!--req, alarm/event triggered or occurred time, format:
2017-07-19T10:06:41+08:00--></dateTime>
    <activePostCount><!--req, xs:integer, alarm/event frequency, starts from 1--
--></activePostCount>
    <eventType><!--req, xs:string, alarm/event type, "peopleCounting, ANPR,..."--
--></eventType>
    <eventState>
        <!--req, xs:string, durative alarm/event status: "active"-valid, "inactive"-
invalid, e.g., when a moving target is detected,
        the alarm/event information will be uploaded continuously unit the status
is set to "inactive"-->
    </eventState>
    <eventDescription><!--req, xs:string, alarm/event description--></
eventDescription>
```

```
<...><!--opt, for different alarm/event types, the nodes are different, see
the message examples in different applications--></...>
</EventNotificationAlert>
```

B.25 XML_EventNotificationAlert_HeartbeatInfo

EventNotificationAlert message with heartbeat information (when there is no alarm is triggered) in XML format

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>
  <portNo><!--opt, xs:integer, device port number--></portNo>
  <protocol><!--opt, xs:string, protocol type for uploading alarm/event
information, "HTTP,HTTPS"--></protocol>
  <macAddress><!--opt, xs:string, MAC address--></macAddress>
  <channelID><!--dep, xs:string, device channel No., starts from 1--></
channelID>
  <dateTime><!--req, heartbeat uploaded time, format:
2017-07-19T10:06:41+08:00--></dateTime>
  <activePostCount><!--req, xs:integer, heartbeat frequency, starts from 1--></
activePostCount>
  <eventType><!--req, xs:string, for heartbeat, it is "videoloss"--></eventType>
  <eventState>
    <!--req, xs:string, for heartbeat, it is "inactive"-->
  </eventState>
  <eventDescription><!--req, xs: string, description--></eventDescription>
</EventNotificationAlert>
```

Remarks

- For network camera or network speed dome with the version 5.5.0 and lower, the heartbeat frequency is 300 ms per heartbeat.
- For network camera or network speed dome with the version 5.5.0 and higher, the heartbeat frequency is 10 s per heartbeat. If no heartbeat received for continuous 30 s, it indicates that the heartbeat is timed out.

Example

Message Example of Heartbeat

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/
XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>videoloss</eventType>
```

```
<eventState>inactive</eventState>
<eventDescription>videoloss alarm</eventDescription>
</EventNotificationAlert>
```

B.26 XML_HttpHostNotification

XML message about parameters of a HTTP listening server

```
<HttpHostNotification version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <url><!--required, xs:string, the absolute path, e.g., http://
<ipAddress><portNo></uri>--></url>
  <protocolType><!--required, xs:string, "HTTP,HTTPS,EHome"--></protocolType>
  <parameterFormatType><!--required, xs:string, alarm/event information format,
"XML,JSON"--></parameterFormatType>
  <addressingFormatType><!--required, xs:string, "ipaddress,hostname"--></
addressingFormatType>
  <hostName><!--dependent, xs:string--></hostName>
  <ipAddress><!--dependent, xs:string--></ipAddress>
  <ipv6Address><!--dependent, xs:string--></ipv6Address>
  <portNo><!--optional, xs:integer--></portNo>
  <userName><!--dependent, xs:string--></userName>
  <password><!--dependent, xs:string--></password>
  <httpAuthenticationMethod><!--required, xs:string, "MD5digest,none"--></
httpAuthenticationMethod>
  <ANPR><!--optional-->
    <detectionUpLoadPicturesType>
      <!--optional, xs:string, types of alarm picture to be uploaded: "all,
licensePlatePicture, detectionPicture"-->
    </detectionUpLoadPicturesType>
  </ANPR>
  <eventType optional="AID,TFS,TPS"><!--required, xs:string--></eventType>
  <uploadImagesDataType>
    <!--optional, xs:string, "URL", "binary" (default), for cloud storage, only
"URL" is supported-->
  </uploadImagesDataType>
  <eventMode><!--optional, xs:string, "all,list"--></eventMode>
  <EventList><!--dependent, it is valid only when eventMode is "list"-->
    <Event><!--required-->
      <type><!--required, xs:string--></type>
    </Event>
  </EventList>
  <channels><!--optional, xs:string, "1,2,3,4..."--></channels>
  <SubscribeEvent/><!--optional, event subscription parameters, see details in
the message of XML_SubscribeEvent-->
</HttpHostNotification>
```

B.27 XML_HttpHostNotificationCap

XML message about capability of HTTP listening server

```
<?xml version="1.0" encoding="utf-8"?>
<HttpHostNotificationCap version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <hostNumber>2</hostNumber>
  <urlLen max=""/>
  <protocolType opt="HTTP,HTTPS,EHome"/>
  <parameterFormatType opt="XML,querystring,JSON"/>
  <addressingFormatType opt="ipaddress,hostname"/>
  <ipAddress opt="ipv4,ipv6"/>
  <portNo min="" max=""/>
  <userNameLen min="" max=""/>
  <passwordLen min="" max=""/>
  <httpAuthenticationMethod opt="MD5digest,none"/>
  <Extensions>
    <intervalBetweenEvents min="" max=""/>
  </Extensions>
  <uploadImagesDataType opt="URL,binary"/>
  <ANPR><!--optional-->
    <detectionUploadPicturesType
opt="all,licensePlatePicture,detectionPicture..."/><!--optional, xs:string, types
of alarm pictures to be uploaded-->
    <alarmHttpPushProtocol opt="baseline,custom"/>
  </ANPR>
  <httpBroken opt="true,false" def="true" ><!--optional, xs:boolean, whether to
enable global ANR: true, false--></httpBroken>
  <SubscribeEventCap>
    <heartbeat min="" max=""/><!--optional, heartbeat time interval, unit:
second-->
    <channelMode opt="all,list"/><!--required, all-subscribe events of all
channels, list-subscribe event by channel-->
    <eventMode opt="all,list"/><!--required, event subscription mode: all-
subscribe all events of all channels, list-subscribe events by type, channel,
and target-->
    <!--if the values of the two nodes channelMode and eventMode are both
"all", it indicates that the device does not support subscribe events by type
and channel-->
    <EventList><!--dependent, alarm uploading mode, this node is valid only
when eventMode is "list"-->
      <Event><!--required-->
        <type><!--required, xs:string, event types--></type>
        <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
        <!--optional, xs:string, transmission format of alarm picture: "binary"-
picture binary data, "localURL"-picture URL from local device,
"cloudStorageURL"-picture URL from cloud storage-->
      </Event>
    </EventList>
    <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
  </SubscribeEventCap>
</HttpHostNotificationCap>
```

```
<!--optional, xs:string, transmission format of all alarm pictures:
"binary"-picture binary data (default for camera), "localURL"-picture URL from
local device (default for NVR/DVR), "cloudStorageURL"-picture URL from cloud
storage; this node is in highest priority-->
<ChangedUploadSub>
  <interval/><!--optional, xs:integer, the life cycle of arming GUID, unit:
second, the default life cycle is 5 minutes; if the reconnection is not started
during the life cycle, a new GUID will be generated-->
  <StatusSub>
    <all/><!--optional, xs:boolean, whether to subscribe all-->
    <channel/><!--optional, xs:boolean, subscribe channel status, this node is
not required when the node all is "true"-->
    <hd/><!--optional, xs:boolean, subscribe the HDD status, this node is not
required when the node all is "true"-->
    <capability/><!--optional, xs:boolean, subscribe the capability changed
status, this node is not required when the node all is "true"-->
  </StatusSub>
</ChangedUploadSub>
</SubscribeEventCap>
</HttpHostNotificationCap>
```

B.28 XML_HttpHostNotificationList

HttpHostNotificationList message in XML format

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <HttpHostNotification>
    <id><!--req, xs:string, ID--></id>
    <url><!--req, xs:string--></url>
    <protocolType><!--req, xs:string, "HTTP,HTTPS"--></protocolType>
    <parameterFormatType><!--req, xs:string, alarm/event information format,
"XML,JSON"--></parameterFormatType>
    <addressingFormatType><!--req, xs:string, "ipaddress,hostname"--></
addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <portNo><!--opt, xs:integer--></portNo>
    <userName><!--dep, xs:string--></userName>
    <password><!--dep, xs:string--></password>
    <httpAuthenticationMethod><!--req, xs:string, "MD5digest,none"--></
httpAuthenticationMethod>
    <uploadImagesDataType>
      <!--opt, xs:string, "URL", "binary" (default), for cloud storage, only
"URL" is supported-->
    </uploadImagesDataType>
    <eventMode><!--opt, xs:string, "all,list"--></eventMode>
    <EventList><!--dep, it is valid only when eventMode is "list"-->
      <Event><!--req-->
```

```
<type><!--req, xs:string--></type>
</Event>
</EventList>
<channels><!--opt, xs:string, "1,2,3,4..."--></channels>
</HttpHostNotification>
</HttpHostNotificationList>
```

Example

HttpHostNotificationList Message Example

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/
XMLSchema">
  <HttpHostNotification>
    <id>1</id>
    <url></url>
    <protocolType>HTTP</protocolType>
    <parameterFormatType>XML</parameterFormatType>
    <addressingFormatType>ipaddress</addressingFormatType>
    <ipAddress>0.0.0.0</ipAddress>
    <portNo>80</portNo>
    <userName></userName>
    <httpAuthenticationMethod>none</httpAuthenticationMethod>
  </HttpHostNotification>
</HttpHostNotificationList>
```

B.29 XML_HttpHostTestResult

HttpHostTestResult message in XML format.

```
<HttpHostTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription>
    <!--req, xs:string-->
  </errorDescription>
</HttpHostTestResult>
```

Appendix C. Response Codes of Text Protocol

The response codes returned during the text protocol integration is based on the status codes of HTTP. 7 kinds of status codes are predefined, including 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid Message Format), 6 (Invalid Message Content), and 7 (Reboot Required). Each kind of status code contains multiple sub status codes, and the response codes are in a one-to-one correspondence with the sub status codes.

StatusCode=1

SubStatusCode	Error Code	Description
ok	0x1	Operation completed.
riskPassword	0x10000002	Risky password.
armProcess	0x10000005	Arming process.

StatusCode=2

Sub Status Code	Error Code	Description
noMemory	0x20000001	Insufficient memory.
serviceUnavailable	0x20000002	The service is not available.
upgrading	0x20000003	Upgrading.
deviceBusy	0x20000004	The device is busy or no response.
reConnectIpc	0x20000005	The video server is reconnected.
transferUpgradePackageFailed	0x20000006	Transmitting device upgrade data failed.
startUpgradeFailed	0x20000007	Starting upgrading device failed.
getUpgradeProcessfailed.	0x20000008	Getting upgrade status failed.
certificateExist	0x2000000B	The Authentication certificate already exists.

StatusCode=3

Sub Status Code	Error Code	Description
deviceError	0x30000001	Hardware error.
badFlash	0x30000002	Flash operation error.
28181Uninitialized	0x30000003	The 28181 configuration is not initialized.
socketConnectError	0x30000005	Connecting to socket failed.
receiveError	0x30000007	Receive response message failed.
deletePictureError	0x3000000A	Deleting picture failed.
pictureSizeExceedLimit	0x3000000C	Too large picture size.
clearCacheError	0x3000000D	Clearing cache failed.
updateDatabasError	0x3000000F	Updating database failed.
searchDatabaseError	0x30000010	Searching in the database failed.
writeDatabaseError	0x30000011	Writing to database failed.
deleteDatabaseError	0x30000012	Deleting database element failed.
searchDatabaseElementError	0x30000013	Getting number of database elements failed.
cloudAutoUpgradeException	0x30000016	Downloading upgrade packet from cloud and upgrading failed.
HBPException	0x30001000	HBP exception.
UDEPException	0x30001001	UDEP exception
elasticSearchException	0x30001002	Elastic exception.
kafkaException	0x30001003	Kafka exception.
HBaseException	0x30001004	Hbase exception.
sparkException	0x30001005	Spark exception.
yarnException	0x30001006	Yarn exception.
cacheException	0x30001007	Cache exception.

Sub Status Code	Error Code	Description
trafficException	0x30001008	Monitoring point big data server exception.
faceException	0x30001009	Human face big data server exception.
SSDFileSystemsIsError	0x30001013	SSD file system error (Error occurs when it is non-Ext4 file system)
insufficientSSDCapacityForFPD	0x30001014	Insufficient SSD space for person frequency detection.
wifiException	0x3000100A	Wi-Fi big data server exception
structException	0x3000100D	Video parameters structure server exception.
noLinkageResource	0x30001015	Insufficient linkage resources.
noArmingResource	0x30001016	Insufficient arming resources.
calibrationTimeout	0x30002051	Calibration timed out.
captureTimeout	0x30006000	Data collection timed out.
lowScore	0x30006001	Low quality of collected data.
uploadingFailed	0x30007004	Uploading failed.

StatusCode=4

Sub Status Code	Error Code	Description
notSupport	0x40000001	Not supported.
lowPrivilege	0x40000002	No permission.
badAuthorization	0x40000003	Authentication failed.
methodNotAllowed	0x40000004	Invalid HTTP method.
notSetHdiskRedund	0x40000005	Setting spare HDD failed.
invalidOperation	0x40000006	Invalid operation.
notActivated	0x40000007	Inactivated.
hasActivated	0x40000008	Activated.
certificateAlreadyExist	0x40000009	The certificate already exists.
operateFailed	0x4000000F	Operation failed.

Sub Status Code	Error Code	Description
USBNotExist	0x40000010	USB device is not connected.
upgradePackageMorethan2GB	0x40001000	Up to 2GB upgrade package is allowed to be uploaded.
IDNotExist	0x40001001	The ID does not exist.
interfaceOperationError	0x40001002	API operation failed.
synchronizationError	0x40001003	Synchronization failed.
synchronizing	0x40001004	Synchronizing.
importError	0x40001005	Importing failed.
importing	0x40001006	Importing.
fileAlreadyExists	0x40001007	The file already exists.
invalidID	0x40001008	Invalid ID.
backupnodeNotAllowLog	0x40001009	Accessing to backup node is not allowed.
exportingError	0x4000100A	Exporting failed.
exporting	0x4000100B	Exporting.
exportEnded	0x4000100C	Exporting stopped.
exported	0x4000100D	Exported.
IPOccupied	0x4000100E	The IP address is already occupied.
IDAlreadyExists	0x4000100F	The ID already exists.
exportItemsExceedLimit	0x40001010	No more items can be exported.
noFiles	0x40001011	The file does not exist.
beingExportedByAnotherUser	0x40001012	Being exported by others.
needReAuthentication	0x40001013	Authentication is needed after upgrade.
unitAddNotOnline	0x40001015	The added data analysis server is offline.
unitControl	0x40001016	The data analysis server is already added.
analysis unitFull	0x40001017	No more data analysis server can be added.
unitIDError	0x40001018	The data analysis server ID does not exist.

Sub Status Code	Error Code	Description
unitExit	0x40001019	The data analysis server already exists in the list.
unitSearch	0x4000101A	Searching data analysis server in the list failed.
unitNotOnline	0x4000101B	The data analysis server is offline.
unitInfoError	0x4000101C	Getting data analysis server information failed.
unitGetNodeInfoError	0x4000101D	Getting node information failed.
unitGetNetworkInfoError	0x4000101E	Getting the network information of data analysis server failed
unitSetNetworkInfoError	0x4000101F	Setting the network information of data analysis server failed
setSmartNodeInfoError	0x40001020	Setting node information failed.
setUnitNetworkInfoError	0x40001021	Setting data analysis server network information failed.
unitRestartCloseError	0x40001022	Rebooting or shutting down data analysis server failed.
virtualIPnotAllowed	0x40001023	Adding virtual IP address is not allowed.
unitInstalled	0x40001024	The data analysis server is already installed.
badSubnetMask	0x40001025	Invalid subnet mask.
uintVersionMismatched	0x40001026	Data analysis server version mismatches.
deviceModelMismatched	0x40001027	Adding failed. Device model mismatches.
unitAddNotSelf	0x40001028	Adding peripherals is not allowed.
noValidUnit	0x40001029	No valid data analysis server.
unitNameDuplicate	0x4000102A	Duplicated data analysis server name.
deleteUnitFirst	0x4000102B	Delete the added data analysis server of the node first.
getLocalInfoFailed	0x4000102C	Getting the server information failed.
getClientAddedNodeFailed	0x4000102D	Getting the added node information of data analysis server failed.
taskExit	0x4000102E	The task already exists.

Sub Status Code	Error Code	Description
taskInitError	0x4000102F	Initializing task failed.
taskSubmitError	0x40001030	Submitting task failed.
taskDelError	0x40001031	Deleting task failed.
taskPauseError	0x40001032	Pausing task failed.
taskContinueError	0x40001033	Starting task failed.
taskSeverNoCfg	0x40001035	Full-text search server is not configured.
taskPicSeverNoCfg	0x40001036	The picture server is not configured.
taskStreamError	0x40001037	Streaming information exception.
taskRecSDK	0x40001038	History recording is not supported.
taskCasaError	0x4000103A	Cascading is not supported.
taskVCARuleError	0x4000103B	Invalid VCA rule.
taskNoRun	0x4000103C	The task is not executed.
unitLinksNoStorageNode	0x4000103D	No node is linked with the data analysis server. Configure the node first.
searchFailed	0x4000103E	Searching video files failed.
searchNull	0x4000103F	No video clip.
userScheOffline	0x40001040	The task scheduler service is offline.
updateTypeUnmatched	0x40001041	The upgrade package type mismatches.
userExist	0x40001043	The user already exists.
userCannotDelAdmin	0x40001044	The administrator cannot be deleted.
userInexistence	0x40001045	The user name does not exist.
userCannotCreatAdmin	0x40001046	The administrator cannot be created.
monitorCamExceed	0x40001048	Up to 3000 cameras can be added.
monitorCunitOverLimit	0x40001049	Adding failed. Up to 5 lower-levels are supported by the control center.
monitorReginOverLimit	0x4000104A	Adding failed. Up to 5 lower-levels are supported by the area.

Sub Status Code	Error Code	Description
monitorArming	0x4000104B	The camera is already armed. Disarm the camera and try again.
monitorSyncCfgNotSet	0x4000104C	The system parameters are not configured.
monitorFdSyncing	0x4000104E	Synchronizing. Try again after completing the synchronization.
monitorParseFailed	0x4000104F	Parsing camera information failed.
monitorCreatRootFailed	0x40001050	Creating resource node failed.
deleteArmingInfo	0x40001051	The camera is already . Disarm the camera and try again.
cannotModify	0x40001052	Editing is not allowed. Select again.
cannotDel	0x40001053	Deletion is not allowed. Select again.
deviceExist	0x40001054	The device already exists.
IPErrorConnectFailed	0x40001056	Connection failed. Check the network port.
cannotAdd	0x40001057	Only the capture cameras can be added.
serverExist	0x40001058	The server already exists.
fullTextParamError	0x40001059	Incorrect full-text search parameters.
storParamError	0x4000105A	Incorrect storage server parameters.
picServerFull	0x4000105B	The storage space of picture storage server is full.
NTPUnconnect	0x4000105C	Connecting to NTP server failed. Check the parameters.
storSerConnectFailed	0x4000105D	Connecting to storage server failed. Check the network port.
storSerLoginFailed	0x4000105E	Logging in to storage server failed. Check the user name and password.
searchSerConnectFailed	0x4000105F	Connecting to full-text search server failed. Check the network port.
searchSerLoginFailed	0x40001060	Logging in to full-text search server failed. Check the user name and password.
kafkaConnectFailed	0x40001061	Connecting to Kafka failed. Check the network port.

Sub Status Code	Error Code	Description
mgmtConnectFailed	0x40001062	Connecting to system failed. Check the network port.
mgmtLoginFailed	0x40001063	Logging in to system failed. Check the user name and password.
TDAConnectFailed	0x40001064	Connecting to traffic data access server failed. Checking the server status.
86sdkConnectFailed	0x40001065	Connecting to listening port of iVMS-8600 System failed. Check the parameters.
nameExist	0x40001066	Duplicated server name.
batchProcessFailed	0x40001067	Processing in batch failed.
IDNotExist	0x40001068	The server ID does not exist.
serviceNumberReachesLimit	0x40001069	No more service can be added.
invalidServiceType.	0x4000106A	Invalid service type.
clusterGetInfo	0x4000106B	Getting cluster group information failed.
clusterDelNode	0x4000106C	Deletion node failed.
clusterAddNode	0x4000106D	Adding node failed.
clusterInstalling	0x4000106E	Creating cluster...Do not operate.
clusterUninstall	0x4000106F	Reseting cluster...Do not operate.
clusterInstall	0x40001070	Creating cluster failed.
clusterIpError	0x40001071	Invalid IP address of task scheduler server.
clusterNotSameSeg	0x40001072	The master node and slave node must be in the same network segment.
clusterVirIpError	0x40001073	Automatically getting virtual IP address failed. Enter manually.
clusterNodeUnadd	0x40001074	The specified master(slave) node is not added.
clusterNodeOffline	0x40001075	The task scheduler server is offline.
nodeNotCurrentIP	0x40001076	The analysis node of the current IP address is required when adding master and slave nodes.
addNodeNetFailed	0x40001077	Adding node failed. The network disconnected.

Sub Status Code	Error Code	Description
needTwoMgmtNode	0x40001078	Two management nodes are required when adding master and slave nodes.
ipConflict	0x40001079	The virtual IP address and data analysis server's IP address conflicted.
ipUsed	0x4000107A	The virtual IP address has been occupied.
cloudAnalyseOnline	0x4000107B	The cloud analytic server is online.
virIP&mainIPnotSame NetSegment	0x4000107C	The virtual IP address is not in the same network segment with the IP address of master/slave node.
getNodeDispatchInfoFailed	0x4000107D	Getting node scheduler information failed.
unableModifyManagementNetworkIP	0x4000107E	Editing management network interface failed. The analysis board is in the cluster.
notSpecifyVirtualIP	0x4000107F	Virtual IP address should be specified for master and slave cluster.
armingFull	0x40001080	No more device can be armed.
armingNoFind	0x40001081	The arming information does not exist.
disArming	0x40001082	Disarming failed.
getArmingError	0x40001084	Getting arming information failed.
refreshArmingError	0x40001085	Refreshing arming information failed.
ArmingPlateSame	0x40001086	The license plate number is repeatedly armed.
ArmingParseXLSError	0x40001087	Parsing arming information file failed.
ArmingTimeError	0x40001088	Invalid arming time period.
ArmingSearchTimeError	0x40001089	Invalid search time period.
armingRelationshipReachesLimit	0x4000108A	No more relation can be created.
duplicateArmingName	0x4000108B	The relation name already exists.
noMoreArmingListAdded	0x4000108C	No more blacklist library can be armed.
noMoreCamerasAdded	0x4000108D	No more camera can be armed.

Sub Status Code	Error Code	Description
noMoreArmingListAddedWithCamera	0x4000108E	No more library can be linked to the camera.
noMoreArmingPeriodAdded	0x4000108F	No more time period can be added to the arming schedule.
armingPeriodsOverlapped	0x40001090	The time periods in the arming schedule are overlapped.
noArmingAlarmInfo	0x40001091	The alarm information does not exist.
armingAlarmUnRead	0x40001092	Getting number of unread alarms failed.
getArmingAlarmError	0x40001093	Getting alarm information failed.
searchByPictureTimedOut	0x40001094	Searching picture by picture timeout. Search again.
comparisonTimeRangeError	0x40001095	Comparison time period error.
selectMonitorNumberUpperLimit	0x40001096	No more monitoring point ID can be filtered.
noMoreComparisonTasksAdded	0x40001097	No more comparison task can be executed at the same time.
GetComparisonResultFailed	0x40001098	Getting comparison result failed.
comparisonTypeError	0x40001099	Comparison type error.
comparisonUnfinished	0x4000109A	The comparison is not completed.
facePictureModelInvalid	0x4000109B	Invalid face model.
duplicateLibraryName.	0x4000109C	The library name already exists.
noRecord	0x4000109D	No record found.
countingRecordsFailed.	0x4000109E	Calculate the number of records failed.
getHumanFaceFrameFailed	0x4000109F	Getting face thumbnail from the picture failed.
modelingFailed.	0x400010A0	Modeling face according to picture URL failed.
1V1FacePictureComparisonFailed	0x400010A1	Comparison 1 VS 1 face picture failed.
libraryArmed	0x400010A2	The blacklist library is armed.

Sub Status Code	Error Code	Description
licenseExceedLimit	0x400010A3	Dongle limited.
licenseExpired	0x400010A4	Dongle expired.
licenseDisabled	0x400010A5	Unavailable dongle.
licenseNotExist	0x400010A6	The dongle does not exist.
SessionExpired	0x400010A7	Session expired .
beyondConcurrentLimit	0x400010A8	Out of concurrent limit.
stopSync	0x400010A9	Synchronization stopped.
getProgressFailed	0x400010AA	Getting progress failed.
uploadExtraCaps	0x400010AB	No more files can be uploaded.
timeRangeError	0x400010AC	Time period error.
dataPortNotConnected	0x400010AD	The data port is not connected.
addClusterNodeFailed	0x400010AE	Adding to the cluster failed. The device is already added to other cluster.
taskNotExist	0x400010AF	The task does not exist.
taskQueryFailed	0x400010B0	Searching task failed.
modifyTimeRuleFailed	0x400010B2	The task already exists. Editing time rule is not allowed.
modifySmartRuleFailed	0x400010B3	The task already exists. Editing VAC rule is not allowed.
queryHistoryVideoFailed	0x400010B4	Searching history video failed.
addDeviceFailed	0x400010B5	Adding device failed.
addVideoFailed	0x400010B6	Adding video files failed.
deleteAllVideoFailed	0x400010B7	Deleting all video files failed.
createVideoIndexFailed	0x400010B8	Indexing video files failed.
videoCheckTypeFailed	0x400010B9	Verifying video files types failed.
configStructuredAddressFailed	0x400010BA	Configuring IP address of structured server failed.
configPictureServerAddressFailed	0x400010BB	Configuring IP address of picture stored server failed.

Sub Status Code	Error Code	Description
storageServiceIPNotExist	0x400010BD	The storage server IP address does not exist.
syncBackupDatabaseFailed	0x400010BE	Synchronizing slave database failed. Try again.
syncBackupNTPTimeFailed	0x400010BF	Synchronizing NTP time of slave server failed.
clusterNotSelectLoopbackAddress	0x400010C0	Loopback address is not supported by the master or slave cluster.
addFaceRecordFailed	0x400010C1	Adding face record failed.
deleteFaceRecordFailed	0x400010C2	Deleting face record failed.
modifyFaceRecordFailed	0x400010C3	Editing face record failed.
queryFaceRecordFailed	0x400010C4	Searching face record failed.
faceDetectFailed	0x400010C5	Detecting face failed.
libraryNotExist	0x400010C6	The library does not exist.
blackListQueryExporting	0x400010C7	Exporting matched blacklists.
blackListQueryExported	0x400010C8	The matched blacklists are exported.
blackListQueryStopExporting	0x400010C9	Exporting matched blacklists is stopped.
blackListAlarmQueryExporting	0x400010CA	Exporting matched blacklist alarms.
blackListAlarmQueryExported	0x400010CB	The matched blacklists alarms are exported.
blackListAlarmQueryStopExporting	0x400010CC	Exporting matched blacklist alarms is stopped.
getBigDataCloudAnalysisFailed	0x400010CD	Getting big data cloud analytic information failed.
setBigDataCloudAnalysisFailed	0x400010CE	Configuring big data cloud analytic failed.

Sub Status Code	Error Code	Description
submitMapSearchFailed	0x400010CF	Submitting search by picture task failed.
controlRelationshipNotExist	0x400010D0	The relation does not exist.
getHistoryAlarmInfoFailed	0x400010D1	Getting history alarm information failed.
getFlowReportFailed	0x400010D2	Getting people counting report failed.
addGuardFailed	0x400010D3	Adding arming configuration failed.
deleteGuardFailed	0x400010D4	Deleting arming configuration failed.
modifyGuardFailed	0x400010D5	Editing arming configuration failed.
queryGuardFailed	0x400010D6	Searching arming configurations failed.
uploadUserSuperCaps	0x400010D7	No more user information can be uploaded.
bigDataServerConnectFailed	0x400010D8	Connecting to big data server failed.
microVideoCloudRequestInfoBuildFailed	0x400010D9	Adding response information of micro video cloud failed.
microVideoCloudResponseInfoBuildFailed	0x400010DA	Parsing response information of micro video cloud failed.
transcodingServerRequestInfoBuildFailed	0x400010DB	Adding response information of transcoding server failed.
transcodingServerResponseInfoParseFailed	0x400010DC	Parsing response information of transcoding server failed.
transcodingServerOffline	0x400010DD	Transcoding server is offline.
microVideoCloudOffline	0x400010DE	Micro video cloud is offline.
UPSServerOffline	0x400010DF	UPS monitor server is offline.
statisticReportRequestInfoBuildFailed	0x400010E0	Adding response information of statistics report failed.
statisticReportResponseInfoParseFailed	0x400010E1	Parsing response information of statistics report failed.
DisplayConfigInfoBuildFailed	0x400010E2	Adding display configuration information failed.

Sub Status Code	Error Code	Description
DisplayConfigInfoParseFailed	0x400010E3	Parsing display configuration information failed.
DisplayConfigInfoSaveFailed	0x400010E4	Saving display configuration information failed.
notSupportDisplayConfigType	0x400010E5	The display configuration type is not supported.
passError	0x400010E7	Incorrect password.
upgradePackageLarge	0x400010EB	Too large upgrade package.
sessionUserReachesLimit	0x400010EC	No more user can log in via session.
ISO8601TimeFormatError	0x400010ED	Invalid ISO8601 time format.
clusterDissolutionFailed	0x400010EE	Deleting cluster failed.
getServiceNodeInfoFailed	0x400010EF	Getting service node information failed.
getUPSInfoFailed	0x400010F0	Getting UPS configuration information failed.
getDataStatisticsReportFailed	0x400010F1	Getting data statistic report failed.
getDisplayConfigInfoFailed	0x400010F2	Getting display configuration failed.
namingAnalysisBoardNotAllowed	0x400010F3	Renaming analysis board is not allowed.
onlyDrawRegionsOfConvexPolygon	0x400010F4	Only drawing convex polygon area is supported.
bigDataServerResponseInfoParseFailed	0x400010F5	Parsing response message of big data service failed.
bigDataServerReturnFailed	0x400010F6	No response is returned by big data service.
microVideoReturnFailed	0x400010F7	No response is returned by micro video cloud service.
transcodingServerReturnFailed	0x400010F8	No response is returned by transcoding service.

Sub Status Code	Error Code	Description
UPSServerReturnFailed	0x400010F9	No response is returned by UPS monitoring service.
forwardingServerReturnFailed	0x400010FA	No response is returned by forwarding service.
storageServerReturnFailed	0x400010FB	No response is returned by storage service.
cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.
modelEmpty	0x400010FD	No model is obtained.
mainAndBackupNodeCannotModifyManagementNetworkInterfaceIP	0x400010FE	Editing the management interface IP address of master node and backup node is not allowed.
IDTooLong	0x400010FF	The ID is too long.
pictureCheckFailed	0x40001100	Detecting picture failed.
pictureModelingFailed	0x40001101	Modeling picture failed.
setCloudAnalysisDefaultProvinceFailed	0x40001102	Setting default province of cloud analytic service failed.
InspectionAreasNumberExceedLimit	0x40001103	No more detection regions can be added.
picturePixelsTooLarge	0x40001105	The picture resolution is too high.
picturePixelsTooSmall	0x40001106	The picture resolution is too low.
storageServiceIPEmpty	0x40001107	The storage server IP address is required.
bigDataServerRequestInfoBuildFail	0x40001108	Creating request message of big data service failed.
analysisTimedOut	0x40001109	Analysis time out.
high-performanceModeDisabled.	0x4000110A	Please enable high-performance mode.
configuringUPSMonitoringServerTimedOut	0x4000110B	Configuring the UPS monitoring server time out. Check IP address.
cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analytic service failed.

Sub Status Code	Error Code	Description
cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analytic service failed.
allCloudAnalysisInterfaceFailed	0x4000110E	Calling API for cloud analytic service failed.
cloudAnalysisModelCompareFailed	0x4000110F	Model comparison of cloud analytic service failed.
cloudAnalysisFacePictureQualityRatingFailed	0x40001110	Getting face quality grading of cloud analytic service failed.
cloudAnalysisExtractFeaturePointsFailed	0x40001111	Extracting feature of cloud analytic service failed.
cloudAnalysisExtractPropertyFailed	0x40001112	Extracting property of cloud analytic service failed.
getAddedNodeInformationFailed	0x40001113	Getting the added nodes information of data analysis server failed.
noMoreAnalysisUnitsAdded	0x40001114	No more data analysis servers can be added.
detectionAreaInvalid	0x40001115	Invalid detection region.
shieldAreaInvalid	0x40001116	Invalid shield region.
noMoreShieldAreasAdded	0x40001117	No more shield region can be drawn.
onlyAreaOfRectangleShapeAllowed	0x40001118	Only drawing rectangle is allowed in detection area.
numberReachedLimit	0x40001119	Number reached the limit.
wait1~3MinutesGetIPAfterSetupDHCP	0x4000111A	Wait 1 to 3 minutes to get IP address after configuring DHCP.
plannedTimeMustbeHalfAnHour	0x4000111B	Schedule must be half an hour.
oneDeviceCannotBuildCluster	0x4000111C	Creating master and backup cluster requires at least two devices.
updatePackageFileNotUploaded	0x4000111E	Upgrade package is not uploaded.

Sub Status Code	Error Code	Description
highPerformanceTasksNotSupportDrawingDetectionRegions	0x4000111F	Drawing detection area is not allowed under high-performance mode.
controlCenterIDDoesNotExist	0x40001120	The control center ID does not exist.
regionIDDoesNotExist	0x40001121	The area ID does not exist.
licensePlateFormatError	0x40001122	Invalid license plate format.
managementNodesNotSupportThisOperation	0x40001123	The operation is not supported.
searchByPictureResourceNotConfiged	0x40001124	The conditions for searching picture by picture are not configured.
videoFileEncapsulationFormatNotSupported	0x40001125	The video container format is not supported.
videoPackageFailure	0x40001126	Converting video container format failed.
videoCodingFormatNotSupported	0x40001127	Video coding format is not supported.
monitorOfDeviceArmingdeleteArmingInfo	0x40001129	The camera is armed. Disarm it and try again.
getVideoSourceTypeFailed	0x4000112A	Getting video source type failed.
smartRulesBuildFailed	0x4000112B	Creating VAC rule failed.
smartRulesParseFailed	0x4000112C	Parsing VAC rule failed.
timeRulesBuildFailed	0x4000112D	Creating time rule failed.
timeRulesParseFailed	0x4000112E	Parsing time rule failed.
monitoInfoInvalid	0x4000112F	Invalid camera information.
addingFailedVersionMismatches	0x40001130	Adding failed. The device version mismatches.
theInformationReturnedAfterCloudAnalysisIsEmpty	0x40001131	No response is returned by the cloud analytic service.

Sub Status Code	Error Code	Description
selectingIpAddressOfHostAndSpareNodeFailedCheckTheStatus	0x40001132	Setting IP address for master node and backup node failed. Check the node status.
theSearchIdDoesNotExist	0x40001133	The search ID does not exist.
theSynchronizationIdDoesNotExist	0x40001134	The synchronization ID does not exist.
theUserIdDoesNotExist	0x40001136	The user ID does not exist.
theIndexCodeDoesNotExist	0x40001138	The index code does not exist.
theControlCenterIdDoesNotExist	0x40001139	The control center ID does not exist.
theAreaIdDoesNotExist	0x4000113A	The area ID does not exist.
theArmingLinkageIdDoesNotExist	0x4000113C	The arming relationship ID does not exist.
theListLibraryIdDoesNotExist	0x4000113D	The list library ID does not exist.
invalidCityCode	0x4000113E	Invalid city code.
synchronizingThePasswordOfSpareServerFailed	0x4000113F	Synchronizing backup system password failed.
editingStreamingTypesNotSupported	0x40001140	Editing streaming type is not supported.
switchingScheduledTaskToTemporaryTaskIsNotSupported	0x40001141	Switching scheduled task to temporary task is not supported.
switchingTemporaryTaskToScheduledTaskIsNotSupported	0x40001142	Switching temporary task to scheduled task is not supported.
theTaskIsNotDispatchedOrItIsUpdating	0x40001143	The task is not dispatched or is updating.
thisTaskDoesNotExist	0x40001144	This task does not exist in the cloud analytic service.
duplicatedSchedule	0x40001145	Schedule period cannot be overlapped.

Sub Status Code	Error Code	Description
continuousScheduleWithSameAlgorithmTypeShouldBeMerged	0x40001146	The continuous schedule periods with same algorithm type should be merged.
invalidStreamingTimeRange	0x40001147	Invalid streaming time period.
invalidListLibraryType	0x40001148	Invalid list library type.
theNumberOfMatchedResultsShouldBeLargerThan0	0x40001149	The number of search results should be larger than 0.
invalidValueRangeOfSimilarity	0x4000114A	Invalid similarity range.
invalidSortingType	0x4000114B	Invalid sorting type.
noMoreListLibraryCanBeLinkedToTheDevice	0x4000114C	No more lists can be added to one device.
InvalidRecipientAddressFormat	0x4000114D	Invalid address format of result receiver.
creatingClusterFailedTheDongleIsNotPluggedIn	0x4000114E	Insert the dongle before creating cluster.
theURLIsTooLong	0x4000114F	No schedule configured for the task.
noScheduleIsConfiguredForTheTask	0x40001150	No schedule configured for the task.
theDongleIsExpired	0x40001151	Dongle has expired.
dongleException	0x40001152	Dongle exception.
invalidKey	0x40001153	Invalid authorization service key.
decryptionFailed	0x40001154	Decrypting authorization service failed.
encryptionFailed	0x40001155	Encrypting authorization service failed.
AuthorizeServiceResponseError	0x40001156	Authorization service response exception.
incorrectParameter	0x40001157	Authorization service parameters error.
operationFailed	0x40001158	Operating authorization service error.
noAnalysisResourceOrNoDataInTheListLibrary	0x40001159	No cloud analytic resources or no data in the list library.

Sub Status Code	Error Code	Description
calculationException	0x4000115A	Calculation exception.
allocatingList	0x4000115B	Allocating list.
thisOperationIsNotSupportedByTheCloudAnalytics	0x4000115C	This operation is not supported by the cloud analytic service.
theCloudAnalyticsIsInterrupted	0x4000115D	The operation of cloud analytic service is interrupted.
theServiceIsNotReady	0x4000115E	The service is not ready.
searchingForExternalApiFailed	0x4000115F	Searching external interfaces failed.
noOnlineNode	0x40001160	No node is online.
noNodeAllocated	0x40001161	No allocated node.
noMatchedList	0x40001162	No matched list.
allocatingFailedTooManyFacePictureLists	0x40001163	Allocation failed. Too many lists of big data service.
searchIsNotCompletedSearchAgain	0x40001164	Current searching is not completed. Search again.
allocatingListIsNotCompleted	0x40001165	Allocating list is not completed.
searchingForCloudAnalyticsResultsFailed	0x40001166	Searching cloud analytic service overtime.
noDataOfTheCurrentLibraryFound	0x40001167	No data in the current library. Make sure there is data in the Hbase.
noFacePictureLibraryIsArmed	0x40001168	No face picture library is armed for big data service.
noAvailableDataSlicingVersionInformationArmedFirstAndSliceTheData	0x40001169	Invalid standard version information.
duplicatedOperationDataSlicingIsExecuting	0x4000116A	Slicing failed. Duplicated operation.
slicingDataFailedNoArmedFacePictureLibrary	0x4000116B	Slicing failed. No arming information in the face big data.

Sub Status Code	Error Code	Description
GenerateBenchmarkFileFailedSlicingAgain	0x4000116C	Generating sliced file failed. Slice again.
NonprimaryNodesProhibitedFromSlicingData	0x4000116D	Slicing is not allowed by the backup node.
NoReadyNodeToClusterServers	0x4000116E	Creating the cluster failed. No ready node.
NodeManagementServicesOffline	0x4000116F	The node management server is offline.
theCamera(s)OfTheControlCenterAreAlreadyArmed.DisarmThemFirst	0x40001170	Some cameras in control center are already armed. Disarm them and try again.
theCamera(s)OfTheAreaAreAlreadyArmed.DisarmThemFirst	0x40001171	Some cameras in this area are already armed. Disarm them and try again.
configuringHigh-frequencyPeopleDetectionFailed	0x40001172	Configuring high frequency people detection failed.
searchingForHigh-frequencyPeopleDetectionLogsFailed.	0x40001173	Searching detection event logs of high-frequency people detection failed.
gettingDetailsOfSearchedHigh-frequencyPeopleDetectionLogsFailed.	0x40001174	Getting the search result details of high frequency alarms failed.
theArmedCamerasAlreadyExistInTheControlCenter	0x40001175	Some cameras in control center are already armed.
disarmingFailedTheCamerasNotArmed	0x40001177	Disarming failed. The camera is not armed.
noDataReturned	0x40001178	No response is returned by the big data service.
preallocFailure	0x40001179	Pre-allocating algorithm resource failed.
overDogLimit	0x4000117A	Configuration failed. No more resources can be pre-allocated.

Sub Status Code	Error Code	Description
analysisServicesDoNotSupport	0x4000117B	Not supported.
commandAndDispatchServiceError	0x4000117C	Scheduling service of cloud analytic service error.
engineModuleError	0x4000117D	Engine module of cloud analytic service error.
streamingServiceError	0x4000117E	Streaming component of cloud analytic service error.
faceAnalysisModuleError	0x4000117F	Face analysis module of cloud analytic service error.
vehicleAnalysisModuleError	0x40001180	Vehicle pictures analytic module of cloud analytic service error.
videoStructuralAnalysisModuleError	0x40001181	Video structuring module of cloud analytic service error.
postprocessingModuleError	0x40001182	Post-processing module of cloud analytic service error.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForListLibrary	0x40001183	High frequency alarm is already armed for blacklist library.
creatingListLibraryFailed	0x40001184	Creating list library failed.
invalidIdentityKeyOfListLibrary	0x40001185	Invalid identity key of list library.
noMoreDevicesCanBeArmed	0x40001186	No more camera can be added.
settingAlgorithmTypeForDeviceFailed	0x40001187	Allocating task resource failed.
gettingHighFrequencyPersonDetectionAlarmInformationFailed	0x40001188	Setting high frequency alarm failed.
invalidSearchCondition	0x40001189	Invalid result.
theTaskIsNotCompleted	0x4000118B	The task is not completed.
resourceOverRemainLimit	0x4000118C	No more resource can be pre-allocated.

Sub Status Code	Error Code	Description
frequentlyAppearedPersonAlarms AlreadyConfiguredForTheCameraDisarmFirstAndTryAgain	0x4000118D	The high frequency alarm of this camera is configured. Delete the arming information and try again.
switchtimedifflesslimit	0x4000123b	Time difference between power on and off should be less than 10 minutes.
associatedFaceLibNumOverLimit	0x40001279	Maximum number of linked face picture libraries reached.
noMorePeopleNumChangeRulesAdded	0x4000128A	Maximum number of people number changing rules reached.
noMoreViolentMotionRulesAdded	0x4000128D	Maximum number of violent motion rules reached.
noMoreLeavePositionRulesAdded	0x4000128E	Maximum number of leaving position rules reached.
SMRDiskNotSupportRaid	0x40001291	SMR disk does not support RAID.
OnlySupportHikAndCustomProtocol	0x400012A3	IPv6 camera can only be added via Device Network SDK or custom protocols.
vehicleEnginesNoResource	0x400012A6	Insufficient vehicle engine resources.
noMoreRunningRulesAdded	0x400012A9	Maximum number of running rules reached.
noMoreGroupRulesAdded	0x400012AA	Maximum number of people gathering rules reached.
noMoreFailDownRulesAdded	0x400012AB	Maximum number of people falling down rules reached.
noMorePlayCellphoneRulesAdded	0x400012AC	Maximum number of playing cellphone rules reached.
ruleEventTypeDuplicate	0x400012C8	Event type duplicated.
noMoreRetentionRulesAdded	0x400015AD	Maximum number of people retention rules reached.

Sub Status Code	Error Code	Description
noMoreSleepOnDutyRulesAdded	0x400015AE	Maximum number of sleeping on duty rules reached.
polygonNotAllowedCrossing	0x400015C2	Polygons are not allowed to cross.
NotAssociatedWithOwnChannel	0x400019C1	Current channel is not linked.
AITargetBPCaptureFail	0x400019C5	Capturing reference picture for AI target comparison failed.
AITargetBPToDSPFail	0x400019C6	Sending reference picture to DSP for AI target comparison failed.
AITargetBPDuplicateName	0x400019C7	Duplicated name of reference picture for AI target comparison.
audioFileNameWrong	0x400019D0	Incorrect audio file name.
audioFileImportFail	0x400019D1	Importing audio file failed.
NonOperationalStandbyMachine	0x400019F0	Non-operational hot spare.
MaximumNumberOfDevices	0x400019F1	The maximum number of devices reached.
StandbyMmachineCannotBeDeleted	0x400019F2	The hot spare cannot be deleted.
alreadyRunning	0x40002026	The application program is running.
notRunning	0x40002027	The application program is stopped.
packNotFound	0x40002028	The software packet does not exist.
alreadyExist	0x40002029	The application program already exists.
noMemory	0x4000202A	Insufficient memory.
invalidLicense	0x4000202B	Invalid License.
noClientCertificate	0x40002036	The client certificate is not installed.
noCACertificate	0x40002037	The CA certificate is not installed.
authenticationFailed	0x40002038	Authenticating certificate failed. Check the certificate.
clientCertificateExpired	0x40002039	The client certificate is expired.

Sub Status Code	Error Code	Description
clientCertificateRevocation	0x4000203A	The client certificate is revoked.
CACertificateExpired	0x4000203B	The CA certificate is expired.
CACertificateRevocation	0x4000203C	The CA certificate is revoked.
connectFail	0x4000203D	Connection failed.
loginNumExceedLimit	0x4000203F	No more user can log in.
HDMIResolutionIllegal	0x40002040	The HDMI video resolution cannot be larger than that of main and sub stream.
hdFormatFail	0x40002049	Formatting HDD failed.
formattingFailed	0x40002056	Formatting HDD failed.
encryptedFormattingFailed	0x40002057	Formatting encrypted HDD failed.
wrongPassword	0x40002058	Verifying password of SD card failed. Incorrect password.
audiosPlayingPleaseWait	0x40002067	Audio is playing. Please wait.
twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress. Please wait.
calibrationPointNumFull	0x40002069	The maximum number of calibration points reached.
completeTheLevelCalibrationFirst	0x4000206A	The level calibration is not set.
completeTheRadarCameraCalibrationFirst	0x4000206B	The radar-camera calibration is not set.
pointsOnStraightLine	0x4000209C	Calibrating failed. The calibration points cannot be one the same line.
TValueLessThanOrEqualZero	0x4000209D	Calibration failed. The T value of the calibration points should be larger than 0.
HBDLibNumOverLimit	0x40002092	The number of human body picture libraries reaches the upper limit

Sub Status Code	Error Code	Description
theShieldRegionError	0x40002093	Saving failed. The shielded area should be the ground area where the shielded object is located.
theDetectionAreaError	0x40002094	Saving failed. The detection area should only cover the ground area.
invalidLaneLine	0x40002096	Saving failed. Invalid lane line.
enableITSFunctionOfThisChannelFirst	0x400020A2	Enable ITS function of this channel first.
noCloudStorageServer	0x400020C5	No cloud storage server
NotSupportWithVideoTask	0x400020F3	This function is not supported.
noDetectionArea	0x400050df	No detection area
armingFailed	0x40008000	Arming failed.
disarmingFailed	0x40008001	Disarming failed.
clearAlarmFailed	0x40008002	Clearing alarm failed.
bypassFailed	0x40008003	Bypass failed.
bypassRecoverFailed	0x40008004	Bypass recovery failed.
outputsOpenFailed	0x40008005	Opening relay failed.
outputsCloseFailed	0x40008006	Closing relay failed.
registerTimeOut	0x40008007	Registering timed out.
registerFailed	0x40008008	Registering failed.
addedByOtherHost	0x40008009	The peripheral is already added by other security control panel.
alreadyAdded	0x4000800A	The peripheral is already added.
armedStatus	0x4000800B	The partition is armed.
bypassStatus	0x4000800C	Bypassed.
zoneNotSupport	0x4000800D	This operation is not supported by the zone.
zoneFault	0x4000800E	The zone is in fault status.
pwdConflict	0x4000800F	Password conflicted.
audioTestEntryFailed	0x40008010	Enabling audio test mode failed.

Sub Status Code	Error Code	Description
audioTestRecoveryFailed	0x40008011	Disabling audio test mode failed.
addCardMode	0x40008012	Adding card mode.
searchMode	0x40008013	Search mode.
addRemoterMode	0x40008014	Adding keyfob mode.
registerMode	0x40008015	Registration mode.
exDevNotExist	0x40008016	The peripheral does not exist.
theNumberOfExDevLimited	0x40008017	No peripheral can be added.
sirenConfigFailed	0x40008018	Setting siren failed.
chanCannotRepeatedBinded	0x40008019	This channel is already linked by the zone.
inProgramMode	0x4000801B	The keypad is in programming mode.
inPaceTest	0x4000801C	In pacing mode.
arming	0x4000801D	Arming.
masterSlavesEnable	0x4000802c	The master-slave relationship has taken effect, the slave radar does not support this operation.
forceTrackNotEnabled	0x4000802d	Mandatory tracking is disabled.
isNotSupportZoneConfigByLocalArea	0x4000802e	This area does not support the zone type.
alarmLineCross	0x4000802f	Trigger lines are overlapped.
zoneDrawingOutOfRange	0x40008030	The drawn zone is out of detection range.
alarmLineDrawingOutOfRange	0x40008031	The drawn alarm trigger line is out of detection range.
hasTargetInWarningArea	0x40008032	The warning zone already contains targets. Whether to enable mandatory arming?
radarModuleConnectFail	0x40008033	Radar module communication failed.
importCfgFilePasswordErr	0x40008034	Incorrect password for importing configuration files.

Sub Status Code	Error Code	Description
overAudioFileNumLimit	0x40008038	The number of audio files exceeds the limit.
audioFileNamesLong	0x40008039	The audio file name is too long.
audioFormatIsWrong	0x4000803a	The audio file format is invalid.
audioFileIsLarge	0x4000803b	The size of the audio file exceeds the limit.
pircamCapTimeOut	0x4000803c	Capturing of pircam timed out.
pircamCapFail	0x4000803d	Capturing of pircam failed.
pircamIsCaping	0x4000803e	The pircam is capturing.
audioFileHasExisted	0x4000803f	The audio file already exists.
subscribeTypeErr	0x4000a016	This metadata type is not supported to be subscribed.
startAppFail	/	Starting running application program failed.
yuvconflict	/	The raw video stream conflicted.
overMaxAppNum	/	No more application program can be uploaded.
noFlash	/	Insufficient flash.
noFlash	/	The platform mismatches.

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Sub Status Code	Error Code	Description
badXmlFormat	0x50000001	Invalid XML format.

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Sub Status Code	Error Code	Description
badParameters	0x60000001	Invalid parameter.
badHostAddress	0x60000002	Invalid host IP address.
badXmlContent	0x60000003	Invalid XML content.
badIPv4Address	0x60000004	Invalid IPv4 address.
badIPv6Address	0x60000005	Invalid IPv6 address.
conflictIPv4Address	0x60000006	IPv4 address conflicted.
conflictIPv6Address	0x60000007	IPv6 address conflicted.

Sub Status Code	Error Code	Description
badDomainName	0x60000008	Invalid domain name.
connectSreverFail	0x60000009	Connecting to server failed.
conflictDomainName	0x6000000A	Domain name conflicted.
badPort	0x6000000B	Port number conflicted.
portError	0x6000000C	Port error.
exportErrorData	0x6000000D	Importing data failed.
badNetMask	0x6000000E	Invalid sub-net mask.
badVersion	0x6000000F	Version mismatches.
badDevType	0x60000010	Device type mismatches.
badLanguage	0x60000011	Language mismatches.
incorrentUserNameOrPasswor d	0x600000012	Incorrect user name or password.
invalidStoragePoolOfCloudServ er	0x600000013	Invalid storage pool. The storage pool is not configured or incorrect ID.
noFreeSpaceOfStoragePool	0x600000014	Storage pool is full.
riskPassword	0x600000015	Risky password.
UnSupportCapture	0x600000016	Capturing in 4096*2160 or 3072*2048 resolution is not supported when H.264+ is enabled.
userPwdLenUnder8	0x600000023	At least two kinds of characters, including digits, letters, and symbols, should be contained in the password.
userPwdNameSame	0x600000025	Duplicated password.
userPwdNameMirror	0x600000026	The password cannot be the reverse order of user name.
beyondARGSRangeLimit	0x600000027	The parameter value is out of limit.
DetectionLineOutofDetectionR egion	0x600000085	The rule line is out of region.

Sub Status Code	Error Code	Description
DetectionRegionError	0x60000086	Rule region error. Make sure the rule region is convex polygon.
DetectionRegionOutOfCountingRegion	0x60000087	The rule region must be marked as red frame.
PedalAreaError	0x60000088	The pedal area must be in the rule region.
DetectionAreaABError	0x60000089	The detection region A and B must be in the a rule frame.
ABRegionCannotIntersect	0x6000008a	Region A and B cannot be overlapped.
customHBPIDError	0x6000008b	Incorrect ID of custom human body picture library
customHBPIDRepeat	0x6000008c	Duplicated ID of custom human body picture library
dataVersionsInHBDLibMismatches	0x6000008d	Database versions mismatches of human body picture library
invalidHBPID	0x6000008e	Invalid human body picture PID
invalidHBDID	0x6000008f	Invalid ID of human body picture library
humanLibraryError	0x60000090	Error of human body picture library
humanLibraryNumError	0x60000091	No more human body picture library can be added
humanImagesNumError	0x60000092	No more human body picture can be added
noHumanInThePicture	0x60000093	Modeling failed, no human body in the picture
analysisEnginesNoResourceError	0x60001000	No analysis engine.
analysisEnginesUsageExcced	0x60001001	The engine usage is overloaded.
PicAnalysisNoResourceError	0x60001002	No analysis engine provided for picture secondary recognition.

Sub Status Code	Error Code	Description
analysisEnginesLoadingError	0x60001003	Initializing analysis engine.
analysisEnginesAbnormaError	0x60001004	Analysis engine exception.
analysisEnginesFacelibImportin g	0x60001005	Importing pictures to face picture library. Failed to edit analysis engine parameters.
analysisEnginesAssociatedChan nel	0x60001006	The analysis engine is linked to channel.
smdEncodingNoResource	0x60001007	Insufficient motion detection encoding resources.
smdDecodingNoResource	0x60001008	Insufficient motion detection decoding resources.
diskError	0x60001009	HDD error.
diskFull	0x6000100a	HDD full.
facelibDataProcessing	0x6000100b	Handling face picture library data.
capturePackageFailed	0x6000100c	Capturing packet failed.
capturePackageProcessing	0x6000100d	Capturing packet.
noSupportWithPlaybackAbstra ct	0x6000100e	This function is not supported. Playback by video synopsis is enabled.
insufficientNetworkBandwidth	0x6000100f	Insufficient network bandwidth.
tapeLibNeedStopArchive	0x60001010	Stop the filing operation of tape library first.
identityKeyError	0x60001011	Incorrect interaction command.
identityKeyMissing	0x60001012	The interaction command is lost.
noSupportWithPersonDensityD etect	0x60001013	This function is not supported. The people density detection is enabled.
ipcResolutionOverflow	0x60001014	The configured resolution of network camera is invalid.

Sub Status Code	Error Code	Description
ipcBitrateOverflow	0x60001015	The configured bit rate of network camera is invalid.
tooGreatTimeDifference	0x60001016	Too large time difference between device and server.
noSupportWithPlayback	0x60001017	This function is not supported. Playback is enabled.
channelNoSupportWithSMD	0x60001018	This function is not supported. Motion detection is enabled.
channelNoSupportWithFD	0x60001019	This function is not supported. Face capture is enabled.
illegalPhoneNumber	0x6000101a	Invalid phone number.
illegalCertificateNumber	0x6000101b	Invalid certificate No.
linkedCameraOutLimit	0x6000101c	Connecting camera timed out.
achieveMaxChannelLimit	0x6000101e	No more channels are allowed.
humanMisInfoFilterEnabledChanNumError	0x6000101f	No more channels are allowed to enable preventing false alarm.
humanEnginesNoResource	0x60001020	Insufficient human body analysis engine resources.
taskNumberOverflow	0x60001021	No more tasks can be added.
collisionTimeOverflow	0x60001022	No more comparison duration can be configured.
invalidTaskID	0x60001023	Invalid task ID.
eventNotSupport	0x60001024	Event subscription is not supported.
invalidEZVIZSecretKey	0x60001034	Invalid verification code for Hik-Connect.
needDoubleVerification	0x60001042	Double verification required
noDoubleVerificationUser	0x60001043	No double verification user
timeSpanNumOverLimit	0x60001044	Max. number of time buckets reached
channelNumOverLimit	0x60001045	Max. number of channels reached

Sub Status Code	Error Code	Description
noSearchIDResource	0x60001046	Insufficient searchID resources
noSupportDeleteStrangerLib	0x60001051	Deleting stranger library is not supported
noSupportCreateStrangerLib	0x60001052	Creating stranger library is not supported
behaviorAnalysisRuleInfoError	0x60001053	Behavior analysis rule parameters error.
safetyHelmetParamError	0x60001054	Hard hat parameters error.
OneChannelOnlyCanBindOneEngine	0x60001077	No more engines can be bound.
engineTypeMismatch	0x60001079	Engine type mismatched.
badUpgradePackage	0x6000107A	Invalid upgrade package.
AudioFileNameDuplicate	0x60001135	Duplicated audio file name.
CurrentAudioFileAIRuleInUseAlreadyDelete	0x60001136	The AI rule linkage related to current audio file has been deleted.
TransitionUseEmmc	0x60002000	Starting device failed. The EMMC is overused.
AdaptiveStreamNotEnabled	0x60002001	The stream self-adaptive function is not enabled.
AdaptiveStreamAndVariableBitrateEnabled	0x60002002	Stream self-adaptive and variable bitrate function cannot be enabled at the same time.
noSafetyHelmetRegion	0x60002023	The hard hat detection area is not configured (if users save their settings without configuring the arming area, they should be prompted to configure one).
unclosedSafetyHelmet	0x60002024	The hard hat detection is enabled (If users save their settings after deleting the arming area, they should be prompted to disable hard hat

Sub Status Code	Error Code	Description
		detection first and then delete the arming area).
width/ heightRatioOfPictureError	0x6000202C	The width/height ratio of the uploaded picture should be in the range from 1:2 to 2:1.
PTZNotInitialized	0x6000202E	PTZ is not initialized.
PTZSelfChecking	0x6000202F	PTZ is self-checking.
PTZLocked	0x60002030	PTZ is locked.
advancedParametersError	0x60002031	Auto-switch interval in advanced parameters cannot be shorter than parking tolerance for illegal parking detection in speed dome rule settings.
resolutionError	0x60005003	Invalid resolution
deployExceedMax	0x60006018	The arming connections exceed the maximum number.
detectorTypeMismatch	0x60008000	The detector type mismatched.
nameExist	0x60008001	The name already exists.
uploadImageSizeError	0x60008016	The size of the uploaded picture is larger than 5 MB.
laneAndRegionOverlap	/	The lanes are overlapped.
unitConfigurationNotInEffect	/	Invalid unit parameter.
ruleAndShieldingMaskConflict	/	The line-rule region overlaps with the shielded area.
wholeRuleInShieldingMask	/	There are complete temperature measurement rules in the shielded area.
LogDiskNotSetReadOnlyInGroupMode	0x60001100	The log HDD in the HDD group cannot be set to read-only.
LogDiskNotSetRedundancyInGroupMode	0x60001101	The log HDD in the HDD group cannot be set to redundancy.

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SubStatusCode	Error Code	Description
rebootRequired	0x70000001	Reboot to take effect.

Appendix D. Error Codes Categorized by Functional Modules

The error codes returned during the text protocol integration is categorized by different functional modules. See the error codes, error descriptions, and debugging suggestions in the table below.

Public Function Module (Error Codes Range: 0x00000000, from 0x00100001 to 0x001fffff)

Error String	Error Code	Description	Debugging Suggestion
success	0x00000000	Succeeded.	
deviceNotActivated	0x00100001	The device is not activated.	Activate the device.
deviceNoPermission	0x00100002	Device operation failed. No permission.	Update user's permission.
deviceNotSupport	0x00100003	This function is not supported.	Check the device capability set and call the API corresponding to supported function.
deviceResourceNotEnough	0x00100004	Insufficient resources.	Release resources.
dataFormatError	0x00100005	Invalid message format.	
resetError	0x00100006	Restoring to factory settings failed. Reactivating device is required after the device is reboot as the Reset button may be stuck.	
parameterError	0x00100007	Incorrect parameter	
	0x00100100	Invalid channel	Check if the channel is valid.
	0x00100101	NPQ live view is not supported for stream encryption.	Replace streaming mode for stream encryption.
	0x00100102	No more channels are allowed for NPQ streaming.	Reduce NPQ streaming channels and try again.
	0x00100103	The stream type is not supported.	Check the requested stream type.

Error String	Error Code	Description	Debugging Suggestion
	0x00100104	The number of connections exceeded limit.	Reduce the number of streaming clients and try again.
	0x00100105	Not enough bandwidth.	Reduce the number of remote streaming channels.

User Function Module (Error Codes Range: from 0x00200001 to 0x002fffff)

Error String	Error Code	Description	Debugging Suggestion
passwordError	0x00200001	Incorrect user name or password.	Check if the password is correct.
userNameNotExist	0x00200002	The account does not exist.	Check if the account exists, or add the account.
userNameLocked	0x00200003	The account is locked.	Wait for the device to unlock.
userNumLimited	0x00200004	The number of users allowed to log in exceeded the upper limit.	Log out.
lowPrivilege	0x00200005	No permissions for this operation	For users operations, check the following situations: • Deleting your own account is not allowed. • Editing your own level or permission is not allowed. • Getting information about users with higher permission is not allowed. • Elevating the user's level or permission is not allowed. For other operations, check according to the following measures: If operations unrelated to user's permission configuration failed, you can check the user type and permission, if not solved, contact the developers.
incorrentUserNameOrPassword	0x00200006	Incorrect user name or password	Check if the configured user name and password are

Error String	Error Code	Description	Debugging Suggestion
			matched. If not, contact the administrator to configure again. If the administrator forgets the password, reset the password of the device.
riskPassword	0x00200007	Risk password	Low password strength. Change password again.
passwordMustContainMorethan8Characters	0x00200008	The password length must be greater than or equal to 8.	Check if the password length is greater than or equal to 8. If not, change password again.
passwordLenNoMoreThan16	0x00200009	The password length cannot be greater than 16.	Check if the password length is greater than 16. If yes, change password again.
adminUserNotAllowedModify	0x0020000a	Editing admin information is not allowed.	Check if the edited account is admin.
confirmPasswordError	0x0020000b	Incorrect confirm password.	Check the confirm password.
passwordMustContainMorethan2Types	0x0020000c	The password must contain at least two or more of followings: numbers, lowercase, uppercase, and special characters.	Check if the configured password conforms the requirements.
passwordContainUserName	0x0020000d	The password cannot contain the user name.	Check if the password contains the user name.
userPwdNameMirror	0x0020000e	The password cannot be reversed user name.	Check if the password is reversed user name.

Time Function Module (Error Codes Range: from 0x00300001 to 0x003fffff)

Error String	Error Code	Description	Debugging Suggestion
manualAdjustmentFailed	0x00300001	Time synchronization failed.	
NTPError	0x00300002	Invalid NTP server address.	Check if the NTP server address is valid.
timeFormatError	0x00300003	Incorrect time format during time calibration.	Incorrect message format or incorrect time format.

Error String	Error Code	Description	Debugging Suggestion
		For example, the time in ISO 8601 format should be "2018-02-01T19:54:04", but the applied time is "2018-02-01 19:54:04".	
beyondTimeRangeLimit	0x00300004	The calibration time is not within the time range supported by the device.	Get the device capability and check if the configured time is within the time range supported by the device.
endtimeEarlierThanBeginTime	0x00300005	The start time of the validity period cannot be later than the end time.	Check if the start time and end time are valid.

Network Function Module (Error Codes Range: from 0x00400001 to 0x004fffff)

Error String	Error Code	Description	Debugging Suggestion
domainNameParseFailed	0x00400001	Parsing domain name failed.	
PPPOEConnectedFailed	0x00400002	Connecting PPPOE to the network failed.	
FTPConnectedFailed	0x00400003	The FTP server is disconnected.	
deviceIPConflicted	0x00400004	IP addresses of devices conflicted.	
libraryConnectedFailed	0x00400005	The image and video library is disconnected.	
fileUploadFailed	0x00400006	Uploading failed.	Check if the network connection is normal. If yes, contact after-sales.
storageServiceDownloadFileFailed	0x00400007	Downloading failed.	Check if the network connection is normal. If yes, contact after-sales.
storageServiceDownloadFileSizeZero	0x00400008	The size of file downloaded from the storage service is 0.	Check if the network connection is normal. If yes, contact after-sales.

Error String	Error Code	Description	Debugging Suggestion
storSerNotConfig	0x00400009	Storage service is not configured.	Check if the configuration is correct.
badHostAddress	0x0040000a	Host address error	Check if the configuration is correct.
badIPv4Address	0x0040000b	Incorrect IPv4 address.	Check if the configuration is correct.
badIPv6Address	0x0040000c	Incorrect IPv6 address.	Check if the configuration is correct.
conflictIPv4Address	0x0040000d	IPv4 address conflict.	Check the configuration status of IPV4 in the network.
conflictIPv6Address	0x0040000e	IPv6 address conflict	Check the configuration status of IPV6 in the network.
badDomainName	0x0040000f	Incorrect domain name.	Check if the configuration is correct.
connectServerFail	0x00400010	Connecting to server failed.	Check if the network is normal and check if the configuration is correct.
conflictDomainName	0x00400011	Domain name conflict.	Check if the configuration is correct.
badPort	0x00400012	Port conflict.	Check if the configuration is correct.
portError	0x00400013	Port error	Check if the configuration is correct.
badNetMask	0x00400014	Subnet mask error	Check if the configuration is correct.
badVersion	0x00400015	Version mismatch	Check if the version is correct.
badDns	0x00400016	DNS error	Check if the configuration is correct.
badMTU	0x00400017	MTU error	Check if the configuration is correct.

Error String	Error Code	Description	Debugging Suggestion
badGateway	0x00400018	Wrong gateway	Check if the configuration is correct.
urlDownloadFail	0x00400019	Downloading via URL failed.	Check if the network is normal and check if the URL is correct.
deployExceedMax	0x0040001a	The number of armed channels exceeds the maximum number of connections.	Get the supported maximum number of arming and the number of armed channels.

Maintenance Function Module (Error Codes Range: from 0x00500001 to 0x005fffff)

Error String	Error Code	Description	Debugging Suggestion
upgradeXMLFormatError	0x00500001	Incorrect XML upgrading request.	Check if the upgrade file is correct. If the file is correct, try the local upgrade.
upgradeContentError	0x00500002	Incorrect upgrading request content.	Check if the upgrade file is correct. If the file is correct, try the local upgrade.
noUpgradePermission	0x00500003	No upgrade permission.	Switch to admin account or ask admin for advanced operation permission.
upgrading	0x00500004	Upgrading...	Wait for the upgrade to complete.
receiveUpgradePackageError	0x00500005	Receiving upgrade package failed.	Check if the network is normal.
upgradePackageLanguageMismatch	0x00500006	Upgrade package language mismatch.	Check the language type of upgrade package and the device.
upgradePackageMismatch	0x00500007	Upgrade file does not match with the device type.	Check the type of upgrade package and device.
OEMCodeMismatch	0x00500008	Upgrade package error. The OEM code mismatch.	Contact after-sales to get the correct upgrade package.
versionMismatch	0x00500009	Upgrade file version mismatch.	Contact after-sales to get the correct upgrade package.

Error String	Error Code	Description	Debugging Suggestion
upgradeHalfFailed	0x0050000c	Error occurred in the halfway of device upgrading. Flash error or cache error.	
deviceParameterImportFailed	0x0050000d	Importing device parameters failed. Device model, version, or platform mismatches.	
deviceEncryptionError	0x0050000e	Upgrade package mismatches. Device encryption error.	
SDCardFormatError	0x00500025	Formatting SD card failed.	
SDCardLoadFailed	0x00500026	Loading page failed after the SD card is inserted.	
NASFailed	0x00500027	Mounting NAS failed.	
hardDiskError	0x00500028	HDD exception (possible reasons: HDD does not exist, incompatible, encrypted, insufficient capacity, formatting exception, array exception, array incompatible, etc.)	
upgradeError	0x00500030	Upgrade error	
upgradePackageSizeMismatch	0x00500032	Mismatch between the actual size of the downloaded upgrade package and the size in the upgrading request.	
upgradePackageSizeExceeded	0x00500033	The size of the package exceeded that of the partition.	
domainNameParseFailedForDownload	0x00500034	Parsing the domain name of the address for downloading failed.	

Error String	Error Code	Description	Debugging Suggestion
netWorkUnstable	0x00500035	Unstable network. Downloading timed out or the maximum number of attempts reached.	
digestValueMismatch	0x00500036	Mismatched digest value.	
signatureVerifyFailed	0x00500037	Verifying the signature failed.	
innerFormatError	0x00500038	Incorrect inner format of the upgrade package.	
memoryNotEnough	0x00500039	Insufficient memory.	
burnFailed	0x0050003a	Burning firmware failed.	
unknownError	0x0050003b	Unknown error occurred in the underlying APIs.	
userCancel	0x0050003c	User requested cancel of current operation.	
systemResume	0x0050003d	Upgrading failed. You can resume via the backup system or minimum system.	
	0x00500080	Upgrade file is not found.	Check if the upgrade package path is too long or if there is a correct upgrade package under the upgrade package path.
	0x00500081	Upgrade file does not match with the engine type.	Select the upgrade package matched with the device engine type.
	0x00500082	Parsing camera domain name failed.	Confirm if the device is correctly configured DNS service and if the camera domain is valid.
	0x00500083	Camera network is unreachable.	Confirm if the local network can access the network where the added channel located.

Live View Module (Error Codes Range: from 0x00600001 to 0x006fffff)

Error String	Error Code	Description	Debugging Suggestion
liveViewFailed	0x00600001	Live view failed. The number of streaming channels exceeded limit.	
	0x00600002	Request packaging format exception.	Check the packaging format of requested live view.
	0x00600003	NPQ will be unavailable after enabling EHome 2.x.	When EHome 2.x is enable, use other live view mode.
	0x00600005	NPQ live view is not supported for channel-zero.	User other live view mode for channel-zero.
	0x00600007	Only virtual stream supports NPQ live view.	Switch to virtual stream.
	0x0060000A	The IP channel is offline.	Check if the IP channel is online and try again.
	0x0060000B	Live view transcoding is not supported by the device.	Use other stream type for live view.
	0x0060000C	Channel-zero is not enabled.	Enable channel-zero before starting live view of channel-zero.
	0x0060000D	Transcoding capability exceeded limit.	Reduce camera resolution or the number of transcoding channels.
	0x00600010	The channel does not have sub-stream.	Use main stream mode for live view.
	0x00600011	NPQ live view is not supported by the device.	Switch to other live view mode.
	0x00600012	NPQ function is disabled.	Enable NPQ function or switch to other live view mode.

Playback Module (Error Codes Range: from 0x00700001 to 0x007fffff)

Error String	Error Code	Description	Debugging Suggestion
	0x00700001	Playback failed. Up to one channel's playback is supported.	
	0x00700002	The speed of playback displayed on video wall is not supported.	Reduce the playback speed.
	0x00700003	The transmission rate of playback stream is too high.	Reduce the transmission rate of playback stream.
	0x00700004	The encoding type of playback stream is not supported.	Provide the stream with encoding type supported by device.
	0x00700005	The container format of playback stream is not supported.	Provide the stream with container format supported by device.
	0x00700007	Exception occurred when decoding playback stream Possible reasons: displaying on video wall exception, image exception, display exception, decoding exception, image is stuck, black screen, invalid stream type, live view is stuck, audio decoding exception, and blurred screen.	
	0x00700008	Playback video does not exit, or searching failed.	Search again or check if HDD is normal.
	0x00700009	Playback time parameter error.	Check if the time period of searched video is correct and try again.
	0x0070000A	Invalid video type.	Select the correct video type to search.
	0x0070000B	Invalid time type.	Select the correct time type to search.

Error String	Error Code	Description	Debugging Suggestion
	0x0070000C	Invalid event parameter.	Select the correct event parameter to search.
	0x0070000D	Invalid event type.	Select the correct event type to search.
	0x0070000E	The device does not support smart search.	Select the non smart search mode to search.
	0x0070000F	Invalid smart event type.	Select the correct smart event type to search.
	0x00700010	Invalid dynamic analysis sensitivity.	Select the correct sensitivity to search video.
	0x00700011	Reverse playback is not supported.	Select the correct playback mode.
	0x00700012	Invalid file status.	Select the correct file status to search.
	0x00700013	Invalid searching start position.	Use the correct searching start position to search.
	0x00700014	Invalid maximum number of searching.	Use the correct maximum number of searching to search.

Capture Module (Error Codes Range: from 0x00800001 to 0x008fffff)

Error String	Error Code	Description	Debugging Suggestion
	0x00800001	Manual capture failed.	

Two-Way Audio Module (Error Codes Range: from 0x00900001 to 0x009fffff)

Error String	Error Code	Description	Debugging Suggestion
startFailed	0x00900001	Starting two-way audio failed. Audio loss or driver error.	
codingFormatNot Match	0x00900002	The encoding format of the intercom is inconsistent, and the negotiation fails	Check or capture the packets on the platform, then analyze if the audio encoding formats negotiated by both sides are consistent.

Error String	Error Code	Description	Debugging Suggestion
dialedIsBusy	0x00900003	The intercom party is already in the intercom and can no longer respond to the intercom	Check if the intercom party is already in the intercom, if not, get the protocol message and analyze the response message.
destinationLongNumberError	0x00900004	The requested destination long number is wrong	Check or capture the packets on the platform, then analyze the long number.

Video Storage Module (Error Codes Range: from 0x00a00001 to 0x00afffff)

Error String	Error Code	Description	Debugging Suggestion
videoSearchFailed	0x00a00001	Searching videos failed.	No resource stored in the device.
notFindStorageMedium	0x00a00002	No storage medium found.	
videoDownloadFailed	0x00a00003	Downloading videos failed.	

Picture Storage Module (Error Codes Range: from 0x00b00001 to 0x00bfffff)

Error String	Error Code	Description	Debugging Suggestion
	0x00b00001	Searching pictures failed.	No picture resource.

IO Function Module (Error Codes Range: from 0x00c00001 to 0x00cfffff)

Error String	Error Code	Description	Debugging Suggestion
	0x00c00001	Invalid alarm input No.	
	0x00c00002	Invalid alarm output No.	

Event Function Module (Error Codes Range: from 0x00d00001 to 0x00dfffff)

Error String	Error Code	Description	Debugging Suggestion
	0x00d00001	Incorrect event rule.	Refer to the manual for correct configuration.

Parking Service Module (Error Codes Range: from 0x00e00001 to 0x00efffff)

Error String	Error Code	Description	Debugging Suggestion
	0x00e00001	The vehicle with parking pass already exists.	Parking pass is created by license plate, you need to check if the parking pass for this license plate already created.
	0x00e00002	The license plate number is required.	

General Function Module (Error Codes Range: from 0x00f00001 to 0x00ffffff)

Error String	Error Code	Description	Debugging Suggestion
noMemory	0x00f00001	Insufficient device memory (heap space allocation failed).	Check the free memory and send logs to the developer for analysis.
deviceBusy	0x00f00002	The device is busy or the device is not responding.	Send logs to the developers for analysis. For fingerprint collection, face collection, file application, and file uploading services, check if the last operation is completed.
notSupport	0x00f00003	The URL is not supported by the device.	Capture the packets, check if the applied URL exists in the PMP platform. If yes, send the URL to the developer for analysis.
methodNotAllowed	0x00f00004	HTTP method is not allowed.	Capture the packets, check the method corresponding to the URL in the PMP platform.
invalidOperation	0x00f00005	Invalid operation of API command.	
IDNotExist	0x00f00006	The ID does not exist (the URL should contain ID, but	Capture the packets and check if the ID included in the URL is correct.

Error String	Error Code	Description	Debugging Suggestion
		the actual URL does not contain the ID).	
invalidID	0x00f00007	Invalid ID (the ID in the URL exceeds the capability set or the ID format is invalid).	Capture the packets and check if the ID included in the URL is correct. Get the capabilities of URL and check the ID range.
invalidIURL	0x00f00008	The content after the "?" in the URL is wrong.	Capture the packets and check if the URL is correct.
deviceAckTimeOut	0x00f00009	Device response timed out.	If the communication with the external module timed out, check if the external module is offline. When the above situation is eliminated, send logs to the developer for analysis.
badXmlFormat	0x00f0000a	XML format error	
badJsonFormat	0x00f0000b	JSON format error	
badURLFormat	0x00f0000c	URL format error	Get the URL and check if it is correct.
badXmlContent	0x00f0000d	XML message error: • The message contains only URL but no message body • The required node is not configured. • Node value exceeds the range limit (incorrect node value).	
badJsonContent	0x00f0000e	JSON message error:	

Error String	Error Code	Description	Debugging Suggestion
		- The message contains only URL but no message body - The required node is not configured. - Node value exceeds the range limit (incorrect node value).	
messageParametersLack	0x00f0000f	The required node does not exist.	
invalidSearchConditions	0x00f00010	Invalid search condition, search again.	Check if searchID is correct.
operObjectNotExist	0x00f00011	The object does not exist (for the operations about door, alarm IO, the object is not added).	Check if door lock is connected.

Door Control Module (Error Codes Range: from 0x01000001 to 0x010fffff)

Error String	Error Code	Description	Debugging Suggestion
multiAuthenticationFailed	0x01000001	Multi-factor authentication status operation failed.	
securityModuleOffline	0x01000002	The safety door control module is offline and fails to open the door.	Check if the safety door control is offline.

Schedule Template Module (Error Codes Range: from 0x01100001 to 0x011fffff)

Error String	Error Code	Description	Debugging Suggestion
planNumberConflict	0x01100001	Plan number conflict.	
timeOverlap	0x01100002	Time period conflict.	Check the message to find out if there is a time overlap of different time periods in one day.

Person Information Module (Error Codes Range: from 0x01200001 to 0x012fffff)

Error String	Error Code	Description	Debugging Suggestion

Certificate Module (Error Codes Range: from 0x01300001 to 0x013fffff)

Error String	Error Code	Description	Debugging Suggestion

Security Function Module (Error Codes Range: from 0x01400001 to 0x014fffff)

Error String	Error Code	Description	Debugging Suggestion
decryptFailed	0x01400001	Decryption failed, when decrypting sensitive information fields or importing data files.	The import secret key should be consistent with the export.
certificateNotmatch	0x01400003	Certificates mismatched, SSL/TLS public and private keys need to be matched in pairs.	The public and private keys need to be generated at the same time.
notActivated	0x01400004	Device is not activated.	Activate the device by tools such as SADP before use.
hasActivated	0x01400005	Device has been activated.	
forbiddenIP	0x01400006	IP address is banned	IP address is banned when illegal login attempts exceed the upper limit.
bondMacAddressNotMatch	0x01400007	The MAC address does not match the user.	Check if the specific MAC address has linked to the user.
bondIpAddressNotMatch	0x01400008	IP address does not match the user.	Check if the specific IP address has linked to the user.
badAuthorization	0x01400009	Triggered by illegal login	Incorrect password triggered the illegal login.

Advertising Function Module (Error Codes Range: from 0x01500001 to 0x015fffff)

Error String	Error Code	Description	Debugging Suggestion
materialDownloadFailed	0x01500001	Material download failed.	• Check if the network connection is normal. • Check if the device is running normally. • Check the log print.
materialNumbersOver	0x01500002	The number of materials in the program list reached the upper limit.	Check if the number of materials in applied program list exceeded the limit.

